CONTRACTIBLE 4-MANIFOLDS WITH NONNEGATIVE RICCI CURVATURE AND EUCLIDEAN VOLUME GROWTH

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Abstract. We prove that every contractible smooth four-manifold admitting a complete Riemannian metric with nonnegative Ricci curvature and Euclidean volume growth is homeomorphic to $\mathbb{R}^4$. More precisely, every compact subset lies in the interior of a connected compact regular domain with simply connected complement. The key construction produces compact domains whose boundaries admit positive scalar curvature and which are regular sublevels of global strictly three-convex Morse exhaustions. A topological criterion forces the exterior fundamental group to be trivial whenever it contains no nonabelian free subgroup and satisfies the resulting boundary, homological, and relative cell hypotheses; in the geometric application, this group is finite. Its proof uses a cellular formulation of the Gerstenhaber–Rothaus theorem and a smooth filling obstruction. The resulting complements give simple connectivity at infinity, and Freedman's recognition theorem completes the argument.

1. Introduction

Contractibility alone does not determine the topology of an open four-manifold. Although every loop contracts in the ambient manifold, loops far from a compact set need not contract while avoiding that set. Controlling this behavior at infinity is therefore central to recognizing a contractible four-manifold as $\mathbb{R}^4$. In this context, Semola asks whether a contractible four-manifold admitting a complete metric with nonnegative Ricci curvature and Euclidean volume growth can fail to be homeomorphic or diffeomorphic to $\mathbb{R}^4$ [29, Question 5.8]. We answer the homeomorphism part.

Condition 1 (Geometric hypotheses)**.** Let M be a smooth four-manifold without boundary, equipped with a smooth Riemannian metric g. Assume that M is contractible, (M, g) is complete, and $\mathrm{Ric}_g \geq 0$. Fix $p \in M$ and assume

$$\mathrm{AVR}(g) := \lim_{R\to\infty} \frac{\mathrm{Vol}_g B_R^g(p)}{\omega_4 R^4} > 0,$$

where ω_4 is the volume of the Euclidean unit ball in $\mathbb{R}^4$. In particular, M is connected and noncompact.

Theorem 1.1. *Under Condition 1, $M \cong_{\mathrm{top}} \mathbb{R}^4$.*

Key words and phrases. Ricci curvature, positive scalar curvature, Morse theory.

More precisely, we obtain the following control of compact complements, writing $O_\Gamma = M \setminus \Gamma$.

Theorem 1.2 (Cofinal simply connected complements)**.** *Under Condition 1, for every compact $Q \subset M$ there is a connected compact regular domain Γ with $Q \subset \operatorname{int}\Gamma$ and simply connected exterior O_Γ.*

Here a compact regular domain is a compact codimension-zero submanifold with nonempty smooth boundary. Theorem 1.2 gives simple connectivity at infinity: loops outside a sufficiently large compact domain can be filled while avoiding any prescribed compact subset. Freedman's recognition theorem then yields Theorem 1.1 [11, Corollary 1.2]. The argument identifies the homeomorphism type of M and does not establish a diffeomorphism with standard $\mathbb{R}^4$.

A related result of Chodosh, Máximo, and Mukherjee [7, Theorem 1.2] shows that a compact contractible smooth four-manifold is homeomorphic to the four-ball if its interior admits a complete metric with uniformly positive scalar curvature. In the present setting, the curvature and volume-growth assumptions supply control of the end without a compactification hypothesis. The main construction produces compact cuts whose boundaries admit positive scalar curvature and which are regular sublevels of global strictly three-convex Morse exhaustions (Theorem 5.12). This allows boundary geometry and exterior topology to be controlled for the same domains.

Theorem 6.14 isolates a criterion for such a domain to have simply connected exterior. Its assumptions, stated in Condition 2, concern the boundary, a relative cell model, homology, and the exterior fundamental group. The group hypothesis only excludes subgroups isomorphic to F_2, and in particular applies to finite or amenable exterior groups. The criterion is independent of the original Ricci curvature and volume-growth hypotheses, and is available for other geometric constructions whenever all of these cut conditions are verified. Thus the argument provides both a recognition result and a criterion that can be applied separately from the analytic construction used here.

After the preliminaries, Sections 3–4 construct the analytic data and establish exterior finiteness. Section 5 constructs the cuts, and Sections 6–7 develop the exterior criterion and prove topological recognition. Detailed proofs appear in the appendices.

2. Preliminaries

Homology and cohomology have integer coefficients unless specified otherwise. We use $\Delta = \sum_i \partial_i^2$ in Euclidean coordinates and the curvature convention for which round spheres have positive sectional curvature.

2.1. Domains and geometric notation.

Definition 2.1 (Compact regular domains)**.** A compact regular domain $K \subset M$ is a compact embedded codimension-zero submanifold with nonempty smooth boundary.

Definition 2.2 (The two exteriors)**.** Set

$$Y_K = \partial K, \qquad O_K = M \setminus K, \qquad E_K = M \setminus \operatorname{int} K.$$

Boundary collars give a homotopy equivalence $O_K \hookrightarrow E_K$ [18].

Definition 2.3 (Differential operators and curvature)**.** For a metric q, use

$$\operatorname{Hess}_q f = \nabla^q df, \qquad \Delta_q f = \operatorname{tr}_q \operatorname{Hess}_q f, \qquad \operatorname{Scal}_q = \operatorname{tr}_q \operatorname{Ric}_q .$$

For a symmetric bilinear form S and a subspace P,

$$\operatorname{tr}^q_P S = \sum_{i=1}^{\dim P} S(e_i, e_i),$$

where (e_i) is a q-orthonormal basis of P. Tensor inequalities are inequalities of quadratic forms.

Definition 2.4 (Mean curvature and regular levels)**.** With ν_q the outward unit normal of Y_K, set

$$\mathrm{II}_q(V, W) = q(\nabla^q_V \nu_q, W), \qquad H_q(Y_K) = \operatorname{tr}^q_{TY_K} \mathrm{II}_q.$$

If $K = \{f \le s\}$ near its boundary and $df \neq 0$ there, then

$$\nu_q = \frac{\nabla_q f}{|df|_q}, \qquad H_q(Y_K) = \frac{\operatorname{tr}^q_{TY_K} \operatorname{Hess}_q f}{|df|_q}.$$

Strict mean convexity means $H_q(Y_K) > 0$.

Definition 2.5 (Positive scalar curvature)**.** A manifold admits PSC if it carries a smooth metric with positive scalar curvature. For Y_K, this metric need not be the induced metric. Uniform positivity means a positive global lower bound.

2.2. **Conformal formulas.** For $\widetilde{q} = e^{2h} q$ in dimension four,

$$|V|_{\widetilde{q}} = e^h |V|_q, \qquad |\alpha|_{\widetilde{q}} = e^{-h} |\alpha|_q,$$

for vectors V and covectors α, and

$$\begin{aligned}
\operatorname{Hess}_{\widetilde{q}} f &= \operatorname{Hess}_q f - dh \otimes df - df \otimes dh + \langle dh, df \rangle_q \, q, \\
\operatorname{Ric}_{\widetilde{q}} &= \operatorname{Ric}_q - 2 \operatorname{Hess}_q h + 2 dh \otimes dh - (\Delta_q h + 2|dh|^2_q) q, \\
\operatorname{Scal}_{\widetilde{q}} &= e^{-2h} (\operatorname{Scal}_q - 6 \Delta_q h - 6 |dh|^2_q).
\end{aligned}$$

For the same outward side of a hypersurface,

$$H_{\widetilde{q}} = e^{-h} (H_q + 3 \nu_q h).$$

These formulas follow the preceding conventions [27].

2.3. **Exhaustions and Morse topology.**

Definition 2.6 (Exhaustions and Morse notation)**.** An exhaustion is a continuous function bounded below with compact sublevel sets. For a smooth function f, write $\operatorname{Crit} f = \{df = 0\}$. It is Morse when its critical Hessians are nondegenerate; $\operatorname{ind}_x f$ denotes the number of negative Hessian eigenvalues.

Definition 2.7 (Three-convexity)**.** For a symmetric form S on a four-dimensional inner product space (V, q), set

$$\lambda^q_{(3)}(S) := \min_{\dim P=3} \operatorname{tr}^q_P S = \lambda^q_1(S) + \lambda^q_2(S) + \lambda^q_3(S),$$

where the eigenvalues are in increasing order. Strict and weak three-convexity of f mean, respectively,

$$\lambda^q_{(3)}(\operatorname{Hess}_q f) > 0, \qquad \lambda^q_{(3)}(\operatorname{Hess}_q f) \geq 0.$$

A strictly three-convex Morse function has indices at most two and strictly mean-convex regular sublevel boundaries.

In a compact four-dimensional Morse band, an index-r critical point contributes an r-handle and, up to homotopy, a relative r-cell. Reversing the band changes the handle index to $4 - r$ [23, 24]. A bound on relative cell dimensions concerns the cells attached to the boundary, not the dimension of the entire model.

2.4. **Volume comparison and Green potentials.**

Definition 2.8 (Volume growth)**.** For a connected complete four-manifold with $\operatorname{Ric}_g \geq 0$,

$$\theta := \operatorname{AVR}(g) = \lim_{R\to\infty} \frac{\operatorname{Vol}_g B^g_R(p)}{\omega_4 R^4}, \qquad \omega_4 = \frac{\pi^2}{2},$$

is independent of p. Euclidean volume growth means $\theta > 0$. Bishop–Gromov comparison [27] then gives

$$\theta\omega_4 r^4 \leq \operatorname{Vol}_g B^g_r(x) \leq \omega_4 r^4 \qquad (x \in M,\ r > 0).$$

Definition 2.9 (Green potentials)**.** A positive Green kernel for $-\Delta_g$ is smooth off the diagonal, locally integrable, and satisfies, for every $\eta \in C^\infty_c(M)$,

$$\int_M G_g(x, y)(-\Delta_g \eta(x))\, d\operatorname{Vol}_g(x) = \eta(y).$$

For a nonnegative source f, write

$$P_f(x) = \int_M G_g(x, y) f(y)\, d\operatorname{Vol}_g(y)$$

where finite [20, 19].

2.5. **Group conventions.** Write F_r for the free group of rank r and $G *_A H$ for the pushout of specified homomorphisms $A \to G$ and $A \to H$.

A group is residually finite if every nonidentity element survives under a homomorphism to a finite group. The regular permutation representation embeds every finite group into a unitary group. Consequently, finite-dimensional unitary representations separate elements of a residually finite group [21, 13].

3. Green potential and a three-convex exhaustion

We construct a Green potential that yields two complete conformal metrics: one with uniformly positive scalar curvature and a weighted Ricci lower bound, and another admitting a strictly three-convex Morse exhaustion. The arguments in this section use only completeness, nonnegative Ricci curvature, and positive asymptotic volume ratio in dimension four; contractibility is not needed.

Proposition 3.1 (Newton potential)**.** *Let (M, g) be a connected complete smooth Riemannian four-manifold without boundary, with $\mathrm{Ric}_g \geq 0$ and $\mathrm{AVR}(g) > 0$. Fix $p \in M$ and put $\rho(x) = 1 + d_g(p, x)$. There exist a smooth positive source f and constants $c, C > 0$ such that*

$$\frac{c}{\rho^3} \leq f \leq \frac{C}{\rho^3}.$$

The minimal positive Green kernel G_g of $-\Delta_g$ exists, and the Newton potential

$$P_f(x) := \int_M G_g(x, y) f(y)\, d\,\mathrm{Vol}_g(y)$$

is finite everywhere and defines a smooth positive function $u = P_f$. There are constants $L, U, D > 0$ such that

$$-\Delta_g u = f, \qquad \frac{L}{\rho} \leq u \leq \frac{U}{\rho}, \qquad |du|_g \leq \frac{D}{\rho^2}. \tag{3.1}$$

These estimates give the following conformal data.

Proposition 3.2 (Conformal data)**.** *Let u be the potential constructed in Proposition 3.1. There are constants $m, B > 0$ such that, with*

$$\varphi = \log u, \qquad \bar{g} = u^2 g = e^{2\varphi} g,$$

the metric $\bar{g}$ is complete, $\sup_M \varphi < \infty$, and $-\varphi$ is an exhaustion. Moreover,

$$\begin{gathered} -\Delta_g u \geq m u^3, \qquad |d\varphi|^2_{\bar{g}} \leq B, \qquad \mathrm{Scal}_{\bar{g}} \geq 6m, \\ \mathrm{Ric}_{\bar{g}} + 2\,\mathrm{Hess}_{\bar{g}}\, \varphi + d\varphi \otimes d\varphi \geq m \bar{g}. \end{gathered} \tag{3.2}$$

The same potential also yields a strictly three-convex Morse exhaustion for a second complete conformal metric. The resulting index bound will control the dimensions of the relative cells in the exterior of a cut.

Theorem 3.3 (Three-convex Morse exhaustion)**.** *For the potential u in Proposition 3.2, set*

$$\psi = \frac{\varphi}{3} = \frac{\log u}{3}, \qquad q = u^{2/3} g = e^{2\psi} g.$$

Then (M, q) is complete and admits a smooth Morse exhaustion F satisfying

$$\lambda^q_{(3)}(\mathrm{Hess}_q F) > 0 \qquad \text{on } M.$$

In particular,

$$\mathrm{ind}_x F \le 2 \qquad (x \in \mathrm{Crit}\, F).$$

The choice of exponent in $q = u^{2/3} g = e^{-4\varphi/3} \bar{g}$ connects the weighted Ricci estimate to three-convexity. To see this, let b be a ray Busemann function for $\bar{g}$, and let h be one of the smooth local lower supports constructed in Lemma A.2. At the contact point, $|dh|_{\bar{g}} = 1$ and $\mathrm{Hess}_{\bar{g}}\, h(\nabla_{\bar{g}} h, \nabla_{\bar{g}} h) = 0$. The conformal Hessian formula therefore gives

$$\mathrm{tr}^q_{\ker dh} \mathrm{Hess}_q\, h = e^{4\varphi/3} \left(\Delta_{\bar{g}} h - 2\langle d\varphi, dh \rangle_{\bar{g}} \right).$$

An index-form estimate using (3.2) makes the expression in parentheses positive. Thus the exponent matches the drift coefficient supplied by the weighted Ricci bound.

Positivity on $\ker dh$ alone does not yet control every three-plane. An increasing convex reparametrization χ contributes the positive term $\chi''(h)\, dh \otimes dh$ to the Hessian; choosing χ''/χ' sufficiently large on successive compact sets gives positivity on all three-planes. The Busemann function is proper by the annular diameter estimate in Lemma A.1. Shen's approximation lemma then produces the smooth strictly three-convex Morse exhaustion [30, Section 3.1, Lemma 5]. The support construction and the locally uniform estimates needed for this approximation are proved in Appendix A.3.

Remark. For the Euclidean model on $\mathbb{R}^4 \setminus \{0\}$, take $u = r^{-1}$, where $r = |x|$. With $t = \log r$ and $s = \frac{3}{2} r^{2/3}$, the conformal metrics become

$$\bar{g} = dt^2 + g_{S^3}, \qquad \mathrm{Scal}_{\bar{g}} = 6, \qquad q = ds^2 + \left(\frac{2s}{3}\right)^2 g_{S^3}.$$

Moreover,

$$\mathrm{Hess}_q\, t = \frac{3}{2s^2}(q - 2\, ds \otimes ds), \qquad \lambda^q_{(3)}(\mathrm{Hess}_q\, t) = \frac{3}{2s^2} > 0.$$

This model illustrates the different roles of the two conformal metrics and is considered only away from the origin.

4. Finite exterior fundamental groups

We retain the complete metric q of Theorem 3.3. Every strictly three-convex Morse exhaustion for q yields a cofinal family of connected mean-convex cores with uniformly finite exterior fundamental groups. The proof combines the annular estimate of [8, Theorem 4.8] with the exterior surjections supplied by the Morse index bound; see Appendix B.

We state the result for an arbitrary exhaustion H, since Section 5 uses both the exhaustion F constructed above and the global defining functions G associated with the chosen cuts.

Theorem 4.1 (Finite exterior fundamental groups)**.** *Under Condition 1, let q be the complete metric fixed above, and let H be any smooth strictly three-convex Morse exhaustion for q. There are strictly increasing regular values $t_i \to \infty$, with $H(p) < t_1$. Let D_i be the component of $\{H \le t_i\}$ containing p. Each D_i is a connected compact regular domain, and both Y_{D_i} and O_{D_i} are path connected. There is a single constant $N_H \in \mathbb{N}$ such that*

$$\begin{aligned} &D_i \subset \operatorname{int} D_{i+1} \quad (i \ge 1), \qquad \bigcup_{i\ge 1} \operatorname{int} D_i = M, \\ &H_q(Y_{D_i}) > 0 \quad (i \ge 1), \\ &|\pi_1(O_{D_i})| \le N_H \quad (i \ge 1), \\ &\pi_1(O_{D_j}) \twoheadrightarrow \pi_1(O_{D_i}) \quad (j \ge i \ge 1). \end{aligned} \tag{4.1}$$

The bound N_H can be chosen to depend only on (M, g), independently of H. The displayed maps are induced by the actual inclusions and compatible basepoint paths.

For the exhaustion F of Theorem 3.3, we write s_i and C_i for these levels and cores; they are fixed from now on.

5. PSC cuts

We construct compact domains whose boundaries admit PSC and which are regular sublevels of global strictly three-convex Morse exhaustions. The identity $\Gamma = \{G \le 0\}$ allows the exterior finiteness theorem to be applied to larger cores of the same exhaustion.

Throughout this section, fix the conformal and Morse data and the compact regular domains C_i from Theorems 3.3–4.1. In particular, the metric q remains fixed.

5.1. Boundary connectedness and fundamental groups.

We first record the boundary theorem used to obtain connectedness and surjectivity on fundamental groups.

Theorem 5.1 (Spectral boundary theorem)**.** *Let (M, g) be a smooth Riemannian four-manifold, and let $K \subset M$ be a connected compact regular domain. Suppose $v > 0$ is smooth near K and*

$$\begin{aligned} (-\Delta_g v)g + v \operatorname{Ric}_g > 0 \quad &\text{on } K, \\ H_g(Y_K) + v^{-1}\nu_g v \ge 0 \quad &\text{on } Y_K. \end{aligned}$$

Then Y_K is connected, and inclusion induces a surjection

$$\pi_1(Y_K) \twoheadrightarrow \pi_1(K).$$

This is [1, Theorem 2, $n = 4$, $\gamma = 1$].

Corollary 5.2 (Mean-convex boundary groups)**.** *For the fixed conformal data, every connected compact regular domain K with $H_q(Y_K) > 0$ has connected boundary, and inclusion induces a surjection*

$$\pi_1(Y_K) \twoheadrightarrow \pi_1(K).$$

Proof. We have

$$(-\Delta_g u)g + u\,\mathrm{Ric}_g \geq mu^3 g > 0.$$

Since $q = u^{2/3}g$, the conformal mean-curvature formula gives

$$H_q(Y_K) = u^{-1/3}\big(H_g(Y_K) + \nu_g \log u\big). \tag{5.1}$$

Apply Theorem 5.1 with $v = u$. □

5.2. **Forcing and an outermost PSC boundary.** The forcing function provides the boundary barriers and the positivity needed to deduce PSC from stability.

Proposition 5.3 (Late positive forcing)**.** *For every i, there are $j > i$ and a positive function $h \in C^\infty(M)$ such that, with*

$$A = C_i, \qquad B = C_j, \qquad k = u^{2/3}h,$$

we have

$$\begin{aligned} A \subset \operatorname{int} B, \qquad & h^2 - 2|dh|_{\bar g} \geq -m, \\ H_q(Y_A) < k, \qquad & H_q(Y_B) > k. \end{aligned} \tag{5.2}$$

Moreover,

$$\mathrm{Scal}_q + \frac{4}{3}k^2 - 2|dk|_q > 0. \tag{5.3}$$

Definition 5.4 (Weakly outermost compact regular domain)**.** Let (M, q) be a smooth Riemannian four-manifold, let B be a compact regular domain, and let $k \in C^\infty(M)$ be positive on B. A compact regular domain $T \subset \operatorname{int} B$ is *weakly outermost for $H_q \leq k$ in B* if $H_q(Y_T) \leq k$ and the following property holds: every compact regular domain L satisfying

$$T \subseteq L \subset \operatorname{int} B, \qquad H_q(Y_L) \leq k$$

must equal T.

We use the following existence theorem both for the PSC boundary and for the hull capture argument below.

Theorem 5.5 (Outermost prescribed-mean-curvature domain)**.** *Let (M, q) be a complete connected oriented Riemannian four-manifold, and let $A \subset \operatorname{int} B$ be compact regular domains. Let $\psi \in C^\infty(M)$ be positive on B, and suppose*

$$H_q(Y_A) < \psi, \qquad H_q(Y_B) > \psi.$$

Then there is a compact regular domain E with

$$A \subset \operatorname{int} E, \qquad E \subset \operatorname{int} B, \qquad H_q(Y_E) = \psi,$$

which contains every compact regular domain D satisfying

$$A \subseteq D \subset \operatorname{int} B, \qquad H_q(Y_D) \leq \psi.$$

This follows from [9, Theorem 5.1] *with initial-data tensor* $(\psi/3)q$*; see Appendix C.*

Proposition 5.6 (Outermost PSC boundary)**.** *For A, B, k in Proposition 5.3, there is a compact regular domain T, weakly outermost for $H_q \leq k$ in B, such that*

$$A \subset \operatorname{int} T, \qquad T \subset \operatorname{int} B, \qquad H_q(Y_T) = k.$$

Its boundary Y_T admits PSC.

Corollary 5.7 (The component containing the inner core)**.** *Let K be the connected component of T containing A. Then K is a compact regular domain with*

$$A \subset \operatorname{int} K, \qquad H_q(Y_K) > 0.$$

The boundary Y_K is connected and admits PSC, and inclusion induces

$$\pi_1(Y_K) \twoheadrightarrow \pi_1(K).$$

Proof. The boundary Y_K is a union of components of Y_T, and $H_q(Y_K) = k > 0$. Apply Corollary 5.2. □

5.3. **Inward collars.** A sufficiently small inward displacement preserves strict mean convexity and the diffeomorphism type of the boundary. We use this freedom in the hull capture argument.

Lemma 5.8 (Inward collar cut)**.** *Let K be the component in Corollary 5.7. There is a connected compact regular domain Γ such that*

$$A \subset \operatorname{int} \Gamma, \qquad \Gamma \subset \operatorname{int} K, \qquad H_q(Y_\Gamma) > 0,$$

and Y_Γ is diffeomorphic to Y_K. Consequently, Y_Γ is connected and admits PSC, and inclusion induces

$$\pi_1(Y_\Gamma) \twoheadrightarrow \pi_1(\Gamma).$$

5.4. **A global defining Morse level for the cut.** For a nonempty compact set K, define its weak three-convex hull by

$$\widehat{K}_3^q = \left\{ x \in M : \begin{array}{l} v(x) \leq \max_K v \text{ for every } v \in C^\infty(M) \\ \text{with } \lambda^q_{(3)}(\operatorname{Hess}_q v) \geq 0 \end{array} \right\}.$$

This is the ambient smooth p-convex hull of [15, Definition 3.1], with $p = 3$ and the Riemannian Hessian convention of [15, Definition 2.2′]. Hull equality means that every point outside K can be separated from K by a smooth weakly three-convex function. It is an ambient separation property; three-convexity of the interior alone does not assert this equality.

The following maximum principle gives a useful way to detect this separation property without constructing the separators individually.

Definition 5.9 (Relative maximum principle)**.** For a closed set S and an open set Ω, write $\mathsf{MP}_q(S;\Omega)$ if every smooth test function ψ having a local maximum on S at $x \in S \cap \Omega$ admits some three-plane $P \subset T_xM$ such that

$$\operatorname{tr}_P^q \operatorname{Hess}_q \psi(x) \le 0.$$

Lemma 5.10 (Direct compact capture)**.** *Let A, B, k, T be the barriers and outermost domain of Proposition 5.6, and let K be its component containing A. Let $\Gamma \subset \operatorname{int} K$ be an inward collar cut. Suppose M has a smooth strictly three-convex exhaustion F, and choose a smooth collar defining function r and a number $c < 0$ satisfying*

$$\Gamma = \{x \in K : r(x) \le c\},$$

and, throughout $K \setminus \Gamma$,

$$dr \neq 0, \qquad \operatorname{tr}^q_{(\nabla_q r)^\perp} \operatorname{Hess}_q r > 0.$$

Every compact $S \subset B$ satisfying $\mathsf{MP}_q(S; M \setminus \Gamma)$ is contained in Γ.

The key point is to fix a protection radius using only the positive lower bound for k, the ambient geometry on B, and the separation of Γ from Y_T; see Lemma C.2. One then enlarges T to contain S and increases the forcing only in a smaller neighborhood of S. The prescribed-mean-curvature theorem gives an enclosing domain E whose boundary stays outside that neighborhood by the same protection estimate. Thus $H_q(Y_E) = k$ on the new boundary, and the original outermostness forces $E = T$. The exhaustion excludes other components of T, and a convex reparametrization of the collar coordinate excludes $S \setminus \Gamma$. The full proof, including the uniform protection estimate, is given in Appendix C.

The weak hull satisfies the relative maximum principle, and the exhaustion makes it compact. For the cuts constructed here, an enclosing regular level of F also places the hull inside B. Lemma 5.10 therefore gives the following ambient hull equality.

Lemma 5.11 (Hull capture for inward cuts)**.** *The inward collar construction in Lemma 5.8 can be made so that*

$$\widehat{\Gamma}_3^q = \Gamma.$$

Together with strict mean convexity, this separation property allows the collar defining function to extend to a global strictly three-convex exhaustion: convex reparametrization makes the local defining function strict, while hull separators and smooth maxima join it to an exhaustion away from the boundary. A fine Morse approximation fixed near the zero level then preserves the chosen domain.

Theorem 5.12 (PSC cuts with global defining Morse levels)**.** *Under Condition 1, retain the fixed data of Theorems 3.3–4.1. For every compact $Q \subset M$, there are a connected compact regular domain Γ and a smooth strictly three-convex Morse exhaustion G for q with*

$$Q \subset \operatorname{int}\Gamma, \qquad H_q(Y_\Gamma) > 0.$$

The boundary Y_Γ is connected and admits PSC, and inclusion induces

$$\pi_1(Y_\Gamma) \twoheadrightarrow \pi_1(\Gamma).$$

The domain and its boundary are given by

$$\Gamma = \{G \le 0\}, \qquad Y_\Gamma = \{G = 0\},$$

with $dG \neq 0$ on Y_Γ. In addition,

$$\begin{aligned} \lambda^q_{(3)}(\operatorname{Hess}_q G) > 0 \quad &\text{on } M, \\ \operatorname{ind}_x G \le 2 \quad &\text{for } x \in \operatorname{Crit} G. \end{aligned}$$

The domain Γ and the function G may depend on Q, while q remains fixed.

The proofs of the hull capture lemma and the global defining-level theorem are given in Appendix C.

6. From a Morse band to a simply connected exterior

This section gives a topological criterion for a PSC cut to have simply connected exterior. A Morse band supplies a relative cell model and, in the geometric application, a finite exterior group. The criterion itself only requires that the exterior group contain no subgroup isomorphic to the free group F_2. The PSC boundary decomposition, a cellular form of the Gerstenhaber–Rothaus theorem, and a smooth filling obstruction then determine the boundary-group map needed for van Kampen.

The criterion is stated in Theorem 6.14 under Condition 2, independently of the geometric construction in Section 5. Detailed proofs are collected in Appendix D.

For a group G, write $H_i(G;\mathbb{Z}) = H_i(BG;\mathbb{Z})$, where BG is a connected CW complex with fundamental group G and contractible universal cover. In particular, $H_1(G;\mathbb{Z}) = G/[G,G]$. For a space X, the groups $H_2(X)$ and $H_2(\pi_1(X);\mathbb{Z})$ need not agree; their comparison below uses the Hopf homology surjection [4, Section 0].

6.1. Topological data supplied by a Morse band.

Definition 6.1 (Mean-convex band Morse data)**.** Let (M, q) be a smooth Riemannian four-manifold, and let $K \subset \operatorname{int} B$ be compact regular domains. Write $\mathsf{Band}_q(K, B)$ if there is $a \in C^\infty(M)$ whose critical points in $B \setminus \operatorname{int} K$

have nondegenerate Hessians. Its boundary values and, for every x in this band, its pointwise conditions must satisfy

$$\begin{aligned} a|_{Y_K} = 0, \qquad a|_{Y_B} = 1, & \\ \operatorname{ind}_x a \le 2 & \quad \text{if } da_x = 0, \\ \operatorname{tr}^q_{(\nabla_q a)^\perp} \operatorname{Hess}_q a(x) > 0 & \quad \text{if } da_x \neq 0. \end{aligned} \tag{6.1}$$

No range restriction is imposed on the interior values of a.

Lemma 6.2 (Boundary orientation of band data)**.** *Suppose* $\mathsf{Band}_q(K, B)$ *and* $H_q(Y_K), H_q(Y_B) > 0$*. Every function* a *realizing these band data satisfies*

$$da \neq 0 \quad \text{on } Y_K \cup Y_B,$$

and $\nabla_q a$ *points outward from both* K *and* B*.*

Definition 6.3 (Relative cell model)**.** Fix a finite CW structure on Y_K. A *relative* $0,1,2$*-cell model* of (E_K, Y_K) consists of a CW complex X, a cell-preserving identification j of Y_K with a subcomplex of X, and a homotopy equivalence $e : X \to E_K$ such that

$$e \circ j = (Y_K \hookrightarrow E_K).$$

The complex X is obtained from $j(Y_K)$ by attaching relative cells of dimensions $0, 1, 2$ and need not be finite.

Proposition 6.4 (Relative topology of a Morse cut)**.** *Under Condition 1, let* H *and* D_i *be as in Theorem 4.1. Suppose* $K \subset \operatorname{int} D_j$ *is a connected compact regular domain with connected boundary,* $H_q(Y_K) > 0$*, and* $\mathsf{Band}_q(K, D_j)$*. Then* O_K *is path connected, inclusion induces a surjection*

$$\pi_1(O_{D_j}) \twoheadrightarrow \pi_1(O_K),$$

and

$$H_\ell(M, K) = 0 \quad (\ell \ge 3), \qquad H_2(M, K) \text{ is torsion-free.} \tag{6.2}$$

Every finite CW structure on Y_K *extends to a relative cell model as in Definition 6.3.*

Corollary 6.5 (Homology and exterior finiteness)**.** *Under the hypotheses of Proposition 6.4,* $H_1(K)$ *is torsion-free and*

$$|\pi_1(O_K)| \le N_H, \qquad H_2(K) = H_3(K) = 0,$$

where N_H *is the bound in Theorem 4.1.*

Proof. The exterior group is a quotient of $\pi_1(O_{D_j})$. Contractibility of M and the exact sequence of (M, K) give

$$\begin{aligned} H_2(K) &\cong H_3(M, K) = 0, \\ H_3(K) &\cong H_4(M, K) = 0, \\ H_1(K) &\cong H_2(M, K). \end{aligned}$$

The assertions follow from Proposition 6.4. □

We collect these consequences with the boundary data needed for the topological argument.

Condition 2 (Topological cut hypotheses)**.** Let M be a contractible smooth four-manifold, and let $K \subset M$ be a connected compact regular domain with connected boundary. Write $Y = Y_K$, $O = O_K$, and $E = E_K$. Assume that Y admits PSC, O is path connected, and $\pi_1(O)$ contains no subgroup isomorphic to F_2. Assume also that $H_1(K)$ is torsion-free and

$$\pi_1(Y) \twoheadrightarrow \pi_1(K), \qquad H_2(K) = H_3(K) = 0. \tag{6.3}$$

The displayed boundary-group map is induced by inclusion. Assume also that the simultaneous boundary data of Proposition 6.7 can be chosen so that the resulting finite CW structure on Y extends to a relative cell model as in Definition 6.3. Only this chosen CW structure is required.

The exclusion of F_2 holds, in particular, for finite groups and for amenable groups. In the geometric application, Corollary 6.5 supplies finiteness, while Proposition 6.4 supplies a relative model for every finite boundary CW structure and hence for the one required here.

Lemma 6.6 (Boundary homology maps)**.** *Let M be a contractible smooth four-manifold, and let $K \subset M$ be a connected compact regular domain with connected boundary. If $H_2(K) = H_3(K) = 0$ and $H_1(K)$ is torsion-free, inclusion induces*

$$\begin{gathered} H_1(Y_K) \xrightarrow{\cong} H_1(K), \qquad H_2(Y_K) \xrightarrow{\cong} H_2(E_K), \\ H_1(E_K) = 0. \end{gathered} \tag{6.4}$$

In particular, $H_1(Y_K)$ is torsion-free.

6.2. **PSC boundary cells and the auxiliary filling.** The next construction records two consequences of the same PSC connected-sum decomposition. An acyclic subcomplex retains the groups of the homology-sphere summands, while boundary surgeries remove the $S^1 \times S^2$ summands. The specified connecting homomorphism in (6.5) relates the homology of the resulting filling to that of K.

Proposition 6.7 (PSC boundary cells and auxiliary filling)**.** *Let K be a connected compact regular domain in a simply connected smooth four-manifold. Suppose $Y = \partial K$ is connected, admits PSC, and $H_1(Y)$ is torsion-free. Let P denote the Poincaré homology sphere.*

The following data can be constructed simultaneously.

(1) A finite CW structure on Y with one three-cell and no higher cells, together with a finite connected acyclic subcomplex

$$A \subset Y^{(2)}, \qquad \dim A \leq 2.$$

There is a selected family of open two-cells $e_1^2, \dots, e_s^2$ whose attaching maps are constant at points of A. Every other two-cell belongs to A. The group $\pi_1(A)$ is residually finite.

(2) *A connected compact regular domain D in an auxiliary smooth four-manifold and a continuous embedding $f : K \hookrightarrow D$ such that, for some integers $r, v \geq 0$,*

$$\partial D \cong \#_{\ell=1}^{v} (\pm P), \qquad \pi_1(\partial D) \cong \pi_1(A),$$
$$\pi_1(Y) \cong \pi_1(\partial D) * F_r, \qquad H_1(Y) \cong \mathbb{Z}^r.$$

Each summand carries one of the two orientations of P, and the empty sum is S^3.

(3) *Identifying K with $f(K)$ when writing the pair (D, K),*

$$H_1(D, K) = 0, \qquad H_2(D, K) \cong \mathbb{Z}^r.$$

For the connecting homomorphism $\delta : H_2(D, K) \to H_1(K)$, there are bases $(e_1, \dots, e_r)$ of $H_2(D, K)$ and $(b_1, \dots, b_r)$ of $H_1(Y)$ such that

$$\delta e_i = (Y \hookrightarrow K)_* b_i \qquad (1 \leq i \leq r). \tag{6.5}$$

The construction uses the PSC classification [7, Proposition 2.2 and Corollary 2.3], finite spines [31, Theorems 2.1 and 2.5], and product-framed boundary surgeries [28, Lemma 8.24 and Example 8.26(i)].

6.3. **Injectivity of the acyclic boundary subcomplex.** The following theorem is a cellular reformulation of the Gerstenhaber–Rothaus theorem [13, Theorem 1(ii)]. The corresponding solvability statement under a vanishing second-homology hypothesis, with solutions in the same unitary group, is also stated in [26, Lemma 2.1(1)]. Residual finiteness converts this extension property into injectivity of the subcomplex group. The exterior criterion combines this extension property with the boundary cells and the smooth filling obstruction.

Theorem 6.8 (A cellular form of the Gerstenhaber–Rothaus theorem)**.** *Let (Z, A) be a finite CW pair, with A and Z connected. Suppose all relative cells have dimensions $0, 1, 2$ and $H_2(Z, A) = 0$. For every integer $d \geq 1$ and every representation $\rho : \pi_1(A) \to U(d)$, there is a representation $\widetilde{\rho} : \pi_1(Z) \to U(d)$ satisfying*

$$\widetilde{\rho} \circ i_* = \rho, \qquad i : A \hookrightarrow Z.$$

If $\pi_1(A)$ is residually finite, then i_* is injective.

Corollary 6.9 (An acyclic subcomplex of a two-complex)**.** *Let Z be a connected, possibly infinite CW complex of dimension at most two, with $H_2(Z) = 0$. Let $A \subset Z$ be a finite connected subcomplex such that $H_1(A) = 0$ and $\pi_1(A)$ is residually finite. Then inclusion induces an injection*

$$\pi_1(A) \hookrightarrow \pi_1(Z).$$

In particular, if $\pi_1(Z)$ is finite, then $\pi_1(A)$ is finite.

The selected sphere cells in Proposition 6.7 account for the second homology of the exterior. Removing them produces the two-complex to which Corollary 6.9 applies.

Lemma 6.10 (The deleted two-complex)**.** *Under Condition 2, choose the boundary CW structure and the simultaneous data of Proposition 6.7 for which the required relative cell model exists. Let A be its acyclic subcomplex. Choose such a relative cell model (X, j, e), and set*

$$Z := X^{(2)} \setminus \bigcup_{\alpha=1}^{s} j(e_\alpha^2).$$

Then Z is a connected CW complex of dimension at most two containing $j(A)$ as a subcomplex, and

$$H_2(Z) = 0, \qquad \pi_1(Z) \cong \pi_1(E_K).$$

Consequently, inclusion and the comparison maps induce an injection

$$\pi_1(A) \hookrightarrow \pi_1(E_K) \cong \pi_1(O_K).$$

6.4. **From the auxiliary filling to a free boundary group.** The remaining obstruction comes from smooth four-dimensional topology. We use the spin condition only through its obstruction $w_2(TD) \in H^2(D; \mathbb{Z}/2)$: an oriented smooth manifold is spin when this class vanishes. We write $\sigma(D)$ for the signature of its intersection form; in particular, $H_2(D; \mathbb{R}) = 0$ implies $\sigma(D) = 0$ [28, Sections 5.2 and 6.4], [10, Sections 2.4 and 3.2].

Theorem 6.11 (Rokhlin)**.** *Let D be a compact smooth spin four-manifold whose boundary is an integral homology sphere Σ. Then*

$$\sigma(D) \equiv 8\mu(\Sigma) \pmod{16},$$

where $\mu(\Sigma) \in \mathbb{Z}/2$ is the Rokhlin invariant. Moreover, $\mu(P) = \mu(-P) = 1$ [10, Sections 3.8–3.9].

Proposition 6.12 (The free boundary group)**.** *Under Condition 2, choose the simultaneous data A, D, r, v of Proposition 6.7 compatible with the relative cell model in that condition. Then $v = 0$, $\partial D \cong S^3$, and*

$$\pi_1(A) = 1, \qquad \pi_1(\partial D) = 1, \qquad \pi_1(Y) \cong F_r, \tag{6.6}$$
$$H_1(D) = H_2(D) = 0.$$

Indeed, $\pi_1(A)$ is a free product of v copies of the binary icosahedral group $\pi_1(P)$ and injects into $\pi_1(O_K)$. Two or more such factors contain F_2, so the group hypothesis forces $v \leq 1$. The auxiliary filling has vanishing first and second homology, and Rokhlin's theorem excludes the remaining case $\partial D = \pm P$. This is the only step at which the restriction on the exterior group is used.

6.5. **From the boundary group to the exterior.** Once the boundary group is free, its inclusion-induced epimorphism onto $\pi_1(K)$ is determined by the homology already established.

Lemma 6.13 (A free epimorphism with trivial homological kernel)**.** *Let $r \geq 0$, and let $a : F_r \twoheadrightarrow G$ be a surjective group homomorphism. If $H_1(a; \mathbb{Z})$ is injective and $H_2(G; \mathbb{Z}) = 0$, then a is an isomorphism.*

Theorem 6.14 (A criterion for simply connected exteriors)**.** *Under Condition 2, O_K is simply connected.*

Proof. By Proposition 6.12, $\pi_1(Y) \cong F_r$. For the inclusion-induced epimorphism

$$a : \pi_1(Y) \twoheadrightarrow \pi_1(K),$$

Lemma 6.6 makes $H_1(a;\mathbb{Z})$ an isomorphism. The Hopf homology surjection [4, Section 0] gives

$$0 = H_2(K) \twoheadrightarrow H_2(\pi_1(K);\mathbb{Z}).$$

Lemma 6.13 therefore shows that a is an isomorphism.

The collared van Kampen pushout now gives

$$1 = \pi_1(M) \cong \pi_1(K) *_{\pi_1(Y)} \pi_1(E_K) \cong \pi_1(E_K). \tag{6.7}$$

Since $O_K \simeq E_K$ and O_K is path connected, O_K is simply connected. □

7. Cofinality and recognition

We now apply the topological criterion to the cuts from Section 5. The defining exhaustion of each cut supplies the Morse band and finite exterior group needed for this application.

Proposition 7.1 (The defining level supplies a Morse band)**.** *Let Γ, G be as in Theorem 5.12, and apply Theorem 4.1 to $H = G$. For a sufficiently large positive level s in the resulting regular sequence, let B be the component of $\{G \le s\}$ containing p. Then*

$$\Gamma \subset \operatorname{int} B, \qquad H_q(Y_B) > 0, \qquad |\pi_1(O_B)| < \infty.$$

Moreover, $\mathsf{Band}_q(\Gamma, B)$ holds and is realized by

$$a = \frac{G}{s}, \qquad a|_{Y_\Gamma} = 0, \qquad a|_{Y_B} = 1.$$

Consequently, Γ satisfies Condition 2.

The proof is given in Appendix E.

Proof of Theorem 1.2. Given a compact set $Q \subset M$, Theorem 5.12 gives Γ, G with $Q \subset \operatorname{int}\Gamma$. Proposition 7.1 verifies Condition 2 for this Γ. Theorem 6.14 therefore makes O_Γ simply connected. □

Definition 7.2 (Simple connectivity at infinity)**.** A noncompact manifold M is *simply connected at infinity* if, for every compact $Q \subset M$, there is a compact $L \supset Q$ such that every loop in $M \setminus L$ extends to a disk in $M \setminus Q$.

Proposition 7.3. *Under Condition 1, M is simply connected at infinity.*

Proof. Given compact Q, choose Γ from Theorem 1.2 and take $L = \Gamma$. Every loop in $M \setminus L = O_\Gamma$ contracts there, hence in $M \setminus Q$. These connected complements form a cofinal family of neighborhoods of infinity, so M also has one end. □

Theorem 7.4 (Contractible open four-manifold recognition)**.** *A contractible open topological four-manifold that is simply connected at infinity is homeomorphic to* $\mathbb{R}^4$ [11, Corollary 1.2, p. 366].

Proof of Theorem 1.1. Under Condition 1, M is an open contractible four-manifold. Proposition 7.3 and Theorem 7.4 therefore give $M \cong_{\mathrm{top}} \mathbb{R}^4$. □

8. Conclusion

The argument separates the geometric construction of cuts from the topological recognition of their exteriors. The analytic construction in Section 3 does not require contractibility. In the contractible setting, Lemma 5.10 gives the compact capture needed to identify an ambient three-convex hull, and Theorem 5.12 realizes the resulting PSC cut as a regular sublevel of a global strictly three-convex Morse exhaustion. This realization makes the boundary geometry and the exterior Morse topology available for the same domain.

Theorem 6.14 can be used independently whenever its boundary, homological, and relative cell hypotheses hold and the exterior group contains no subgroup isomorphic to F_2. It therefore includes finite and amenable exterior groups. The cellular Gerstenhaber–Rothaus theorem supplies the algebraic input; its combination with the boundary cell structure and the smooth filling obstruction is what yields the exterior criterion. The recognition obtained here is topological. A diffeomorphism conclusion would require additional control of the smooth structure.

References

[1] G. Antonelli, Y. Li, P. Sweeney Jr., *Rigidity and flexibility under spectral Ricci lower bounds and mean-convex boundary*, arXiv:2605.11384v1 (2026). doi:10.48550/arXiv.2605.11384 (arXiv DOI).

[2] D. Azagra, J. Ferrera, F. López-Mesas, Y. Rangel, *Smooth approximation of Lipschitz functions on Riemannian manifolds*, J. Math. Anal. Appl. **326** (2007), no. 2, 1370–1378. doi:10.1016/j.jmaa.2006.03.088. arXiv:math/0602051.

[3] F. Berlai, *Residual properties of free products*, Comm. Algebra **44** (2016), no. 7, 2959–2980. doi:10.1080/00927872.2015.1065858. arXiv:1405.0244v2.

[4] K. S. Brown, *Lectures on the Cohomology of Groups*, in *Cohomology of Groups and Algebraic K-Theory*, Adv. Lect. Math. (ALM), vol. 12, International Press, Somerville, MA, 2010, pp. 131–166. Author's version.

[5] M. Brown, *Locally flat imbeddings of topological manifolds*, Ann. of Math. (2) **75** (1962), no. 2, 331–341. doi:10.2307/1970177. Full text.

[6] J. S. Calcut, R. E. Gompf, *On uniqueness of end sums and 1-handles at infinity*, Algebr. Geom. Topol. **19** (2019), no. 3, 1299–1339. doi:10.2140/agt.2019.19.1299. arXiv:1801.00032v2.

[7] O. Chodosh, D. Máximo, A. Mukherjee, *Complete Riemannian 4-manifolds with uniformly positive scalar curvature*, arXiv:2407.05574v1 (2024). doi:10.48550/arXiv.2407.05574 (arXiv DOI).

[8] A. Cucinotta, M. Magnabosco, D. Semola, *New topological restrictions for spaces with nonnegative Ricci curvature*, arXiv:2601.14231v1 (2026). doi:10.48550/arXiv.2601.14231 (arXiv DOI).

[9] M. Eichmair, *Existence, regularity, and properties of generalized apparent horizons*, Comm. Math. Phys. **294** (2010), no. 3, 745–760. doi:10.1007/s00220-009-0970-6. arXiv:0805.4454v3.

[10] S. Finashin, V. Kharlamov, *A glimpse into Rokhlin's Signature Divisibility Theorem*, arXiv:2012.06389v1 (2020). doi:10.48550/arXiv.2012.06389 (arXiv DOI).

[11] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), no. 3, 357–453. doi:10.4310/jdg/1214437136.

[12] G. J. Galloway, R. Schoen, *A generalization of Hawking's black hole topology theorem to higher dimensions*, Comm. Math. Phys. **266** (2006), no. 2, 571–576. doi:10.1007/s00220-006-0019-z. arXiv:gr-qc/0509107.

[13] M. Gerstenhaber, O. S. Rothaus, *The solution of sets of equations in groups*, Proc. Natl. Acad. Sci. USA **48** (1962), no. 9, 1531–1533. doi:10.1073/pnas.48.9.1531.

[14] D. Gilbarg, N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, 2nd ed., Classics in Mathematics, Springer, 2001 (reprint of the 1998 edition). doi:10.1007/978-3-642-61798-0 (electronic edition).

[15] F. R. Harvey, H. B. Lawson, Jr., *p-convexity, p-plurisubharmonicity, and the Levi problem*, Indiana Univ. Math. J. **62** (2013), no. 1, 149–169. doi:10.1512/iumj.2013.62.4886. arXiv:1111.3895v2.

[16] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Author's online version.

[17] M. W. Hirsch, *Differential Topology*, Graduate Texts in Mathematics, vol. 33, Springer, 1976. doi:10.1007/978-1-4684-9449-5.

[18] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Graduate Texts in Mathematics, vol. 218, Springer, 2012. doi:10.1007/978-1-4419-9982-5.

[19] P. Li, L.-F. Tam, *Symmetric Green's functions on complete manifolds*, Amer. J. Math. **109** (1987), no. 6, 1129–1154. doi:10.2307/2374588.

[20] P. Li, S.-T. Yau, *On the parabolic kernel of the Schrödinger operator*, Acta Math. **156** (1986), 153–201. doi:10.1007/BF02399203.

[21] R. C. Lyndon, P. E. Schupp, *Combinatorial Group Theory*, Classics in Mathematics, Springer, 2001 (reprint of the 1977 edition). doi:10.1007/978-3-642-61896-3.

[22] J. Malestein, A. Putman, *On the self-intersections of curves deep in the lower central series of a surface group*, Geom. Dedicata **149** (2010), no. 1, 73–84. doi:10.1007/s10711-010-9465-z. arXiv:0901.2561v3.

[23] J. Milnor, *Morse Theory*, Annals of Mathematics Studies, vol. 51, Princeton University Press, 1963. doi:10.1515/9781400881802 (electronic edition).

[24] J. Milnor, *Lectures on the h-Cobordism Theorem*, Mathematical Notes, vol. 1, Princeton University Press, 1965. doi:10.1515/9781400878055 (electronic edition). 1965 edition scan.

[25] L. Ni, Y. Shi, L.-F. Tam, *Poisson equation, Poincaré–Lelong equation and curvature decay on complete Kähler manifolds*, J. Differential Geom. **57** (2001), no. 2, 339–388. doi:10.4310/jdg/1090348114. Full text.

[26] M. Nitsche, A. Thom, *Universal solvability of group equations*, J. Group Theory **25** (2022), no. 1, 1–10. doi:10.1515/jgth-2019-0167. arXiv:1811.07737v2.

[27] P. Petersen, *Riemannian Geometry*, 3rd ed., Graduate Texts in Mathematics, vol. 171, Springer, 2016. doi:10.1007/978-3-319-26654-1.

[28] A. Ranicki, *Algebraic and Geometric Surgery*, Oxford Mathematical Monographs, Oxford University Press, 2002. Author's corrected electronic version (March 2014).

[29] D. Semola, *The large scale structure of complete 4-manifolds with nonnegative Ricci curvature and Euclidean volume growth*, Indag. Math. (N.S.) **37** (2026), no. 3, 764–798. doi:10.1016/j.indag.2025.04.003. Cited version: arXiv:2501.07125v1 (2025).

[30] Z. Shen, *On complete manifolds of nonnegative kth-Ricci curvature*, Trans. Amer. Math. Soc. **338** (1993), no. 1, 289–310. doi:10.1090/S0002-9947-1993-1112548-6.

[31] A. Yu. Vesnin, S. V. Matveev, E. A. Fominykh, *New aspects of complexity theory for 3-manifolds*, Russian Math. Surveys **73** (2018), no. 4, 615–660. doi:10.1070/RM9829.

Appendix

Appendix A. Proofs of Section 3

Throughout this appendix, (M, g) and p satisfy the hypotheses of Proposition 3.1; no contractibility assumption is made.

A.1. **Proof of Proposition 3.1.**

Proof. We first construct a smooth positive source f satisfying

$$c\rho^{-3} \le f \le C\rho^{-3}.$$

Since M is second-countable and d_g induces its manifold topology, (M, d_g) is separable. Moreover,

$$|\rho(x)^{-3} - \rho(y)^{-3}| \le 3|\rho(x) - \rho(y)| \le 3d_g(x, y),$$

so ρ^{-3} is Lipschitz.

Fix $0 < \varepsilon < 1$. Applying [2, Theorem 1] to ρ^{-3} with the positive continuous error $\varepsilon\rho^{-3}$, we obtain $f \in C^\infty(M)$ such that

$$|f - \rho^{-3}| \le \varepsilon\rho^{-3}.$$

Consequently,

$$0 < (1 - \varepsilon)\rho^{-3} \le f \le (1 + \varepsilon)\rho^{-3}$$

by setting $c = 1 - \varepsilon$ and $C = 1 + \varepsilon$.

Write $\theta = \mathrm{AVR}(g) > 0$. For every $x \in M$ and $t > 0$, Bishop–Gromov comparison [27] gives

$$\theta\omega_4 t^4 \le \mathrm{Vol}_g(B_t^g(x)) \le \omega_4 t^4.$$

Let H_g be the heat kernel obtained by Dirichlet exhaustion, and fix $0 < \eta < 1$. By [20, Corollary 3.1], there is $A > 0$ such that

$$H_g(x, y, t) \le \frac{A}{\mathrm{Vol}_g(B^g_{\sqrt{t}}(x))} \exp\left(-\frac{d_g(x, y)^2}{(4 + \eta)t}\right).$$

Consequently, for $x \ne y$,

$$\begin{aligned} G_g(x, y) &:= \int_0^\infty H_g(x, y, t)\, dt \\ &\le \frac{A}{\theta\omega_4} \int_0^\infty t^{-2} \exp\left(-\frac{d_g(x, y)^2}{(4 + \eta)t}\right) dt \\ &= \frac{A(4 + \eta)}{\theta\omega_4 d_g(x, y)^2} < \infty. \end{aligned}$$

The heat-kernel representation therefore gives a positive symmetric Green kernel for $-\Delta_g$, agreeing with the Dirichlet exhaustion construction [20, Section 5].

Now [20, Theorem 5.2], together with the volume bounds, yields

$$G_g(x,y) \geq C_{10} \int_{d_g(x,y)^2}^{\infty} \frac{dt}{\mathrm{Vol}_g(B^g_{\sqrt{t}}(x))} \geq \frac{C_{10}}{\omega_4 d_g(x,y)^2},$$
$$G_g(x,y) \leq C_{11} \int_{d_g(x,y)^2}^{\infty} \frac{dt}{\mathrm{Vol}_g(B^g_{\sqrt{t}}(x))} \leq \frac{C_{11}}{\theta\omega_4 d_g(x,y)^2}.$$

Thus

$$a\, d_g(x,y)^{-2} \leq G_g(x,y) \leq b\, d_g(x,y)^{-2}, \qquad x \neq y,$$

where $a = C_{10}/\omega_4$ and $b = C_{11}/(\theta\omega_4)$.

Define the source averages

$$k_f(x,t) = \frac{1}{\mathrm{Vol}_g(B^g_t(x))} \int_{B^g_t(x)} f(y)\, d\,\mathrm{Vol}_g(y).$$

Polar-coordinate volume comparison yields, for every $s > 0$,

$$\int_{B^g_s(p)} f(y)\, d\,\mathrm{Vol}_g(y) \leq 4C\omega_4 \int_0^s \frac{r^3}{(1+r)^3}\, dr \leq 4C\omega_4 s.$$

If $t \leq \rho(x)/2$, then $\rho(y) \geq \rho(x)/2$ on $B^g_t(x)$. If $t > \rho(x)/2$, then $B^g_t(x) \subset B^g_{3t}(p)$. Consequently,

$$k_f(x,t) \leq \begin{cases} 8C\rho(x)^{-3}, & 0 < t \leq \rho(x)/2, \\ 12C\theta^{-1}t^{-3}, & t > \rho(x)/2. \end{cases}$$

Integrating these two ranges gives

$$\begin{aligned} \int_0^{\infty} t\, k_f(x,t)\, dt &\leq \frac{C(1+24/\theta)}{\rho(x)}, \\ \int_0^{\infty} k_f(x,t)\, dt &\leq \frac{C(4+24/\theta)}{\rho(x)^2}. \end{aligned} \tag{A.1}$$

Set

$$P_f(x) = \int_M G_g(x,y) f(y)\, d\,\mathrm{Vol}_g(y).$$

By Tonelli's theorem and $d_g(x,y)^{-2} = 2\int_{d_g(x,y)}^{\infty} t^{-3}\, dt$,

$$\begin{aligned} P_f(x) &\leq 2b \int_0^{\infty} t^{-3} \int_{B^g_t(x)} f(y)\, d\,\mathrm{Vol}_g(y)\, dt \\ &\leq 2b\omega_4 \int_0^{\infty} t\, k_f(x,t)\, dt \leq \frac{U}{\rho(x)} \end{aligned}$$

for some $U > 0$. In particular, P_f is finite everywhere.

For the lower bound, on $B^g_{\rho(x)}(x)$ we have $d_g(x,y) < \rho(x)$ and $\rho(y) < 2\rho(x)$. Hence

$$P_f(x) \geq \frac{ac}{8\rho(x)^5}\, \mathrm{Vol}_g(B^g_{\rho(x)}(x)) \geq \frac{ac\theta\omega_4}{8\rho(x)} = \frac{L}{\rho(x)}.$$

The Dirichlet exhaustion kernel G_g is the minimal positive Green kernel; in particular, M is nonparabolic. The volume and Green bounds give, for some $\sigma > 0$,

$$\sigma^{-1}\frac{d_g(x,y)^2}{\mathrm{Vol}_g(B^g_{d_g(x,y)}(x))} \le G_g(x,y) \le \sigma\frac{d_g(x,y)^2}{\mathrm{Vol}_g(B^g_{d_g(x,y)}(x))}, \qquad x \neq y.$$

Together with $\int_0^\infty k_f(p,t)\,dt < \infty$ from (A.1), these verify the hypotheses of [25, Theorem 1.1].

Let w be the solution constructed there, satisfying $\Delta_g w = f$ and $w(p) = 0$. Since

$$\int_M |G_g(p,y) - G_g(x,y)|f(y)\,d\,\mathrm{Vol}_g(y) \le P_f(p) + P_f(x) < \infty,$$

the Green representation preceding [25, equation (1.13)] gives

$$\begin{aligned} w(x) &= w(x) - w(p) \\ &= \int_M (G_g(p,y) - G_g(x,y))\,f(y)\,d\,\mathrm{Vol}_g(y) \\ &= P_f(p) - P_f(x). \end{aligned}$$

Thus $u = P_f(p) - w = P_f$ satisfies $-\Delta_g u = f$. Since f and g are smooth, interior elliptic regularity [14, Theorem 6.17] gives $u \in C^\infty(M)$. The preceding estimates yield

$$\frac{L}{\rho} \le u \le \frac{U}{\rho},$$

so u is positive.

Finally, [25, Theorem 1.3(i)] gives a constant $K = K(4,\sigma) > 0$ such that

$$|du|_g(x) = |dw|_g(x) \le K\int_0^\infty k_f(x,t)\,dt \le \frac{KC(4+24/\theta)}{\rho(x)^2},$$

where the last inequality follows from (A.1). Taking $D = KC(4+24/\theta)$ proves (3.1). □

A.2. **Proof of Proposition 3.2.**

Proof. Retain the potential and constants from Proposition 3.1. Set

$$m = \frac{c}{U^3}, \qquad B = \frac{D^2}{L^4}, \qquad \varphi = \log u, \qquad \bar g = e^{2\varphi}g.$$

Then

$$-\Delta_g u = f \ge \frac{c}{\rho^3} \ge mu^3.$$

We use the conformal identities recalled in Section 2. For $\widetilde g = e^{2\eta}g$ and a smooth function h, they give

$$\mathrm{Hess}_{\widetilde g} h = \mathrm{Hess}_g h - d\eta\otimes dh - dh\otimes d\eta + \langle d\eta, dh\rangle_g\, g. \tag{A.2}$$

In dimension four,

$$\mathrm{Ric}_{\widetilde g} = \mathrm{Ric}_g - 2\,\mathrm{Hess}_g\,\eta + 2d\eta\otimes d\eta - (\Delta_g\eta + 2|d\eta|_g^2)g, \tag{A.3}$$

and

$$\mathrm{Scal}_{\tilde{g}} = e^{-2\eta}(\mathrm{Scal}_g - 6\Delta_g\eta - 6|d\eta|_g^2). \tag{A.4}$$

For $\bar{g} = u^2 g$ we take $\eta = \varphi = \log u$. Differentiation gives

$$\begin{aligned} d\varphi &= u^{-1}du, \\ \mathrm{Hess}_g \varphi &= u^{-1}\,\mathrm{Hess}_g u - u^{-2}du\otimes du, \\ \Delta_g\varphi &= u^{-1}\Delta_g u - u^{-2}|du|_g^2 = -\frac{f}{u} - u^{-2}|du|_g^2, \\ |d\varphi|_g^2 &= u^{-2}|du|_g^2. \end{aligned} \tag{A.5}$$

Substituting $\eta = \varphi$ in (A.3) and taking $h = \varphi$ in (A.2), we obtain

$$\begin{aligned} \mathrm{Ric}_{\bar{g}} &= \mathrm{Ric}_g - 2\,\mathrm{Hess}_g\varphi + 2d\varphi\otimes d\varphi - (\Delta_g\varphi + 2|d\varphi|_g^2)g \\ &= \mathrm{Ric}_g - \frac{2}{u}\,\mathrm{Hess}_g u + \frac{4}{u^2}du\otimes du - \left(\frac{\Delta_g u}{u} + \frac{|du|_g^2}{u^2}\right)g, \\ \mathrm{Hess}_{\bar{g}}\varphi &= \mathrm{Hess}_g\varphi - 2d\varphi\otimes d\varphi + |d\varphi|_g^2 g \\ &= u^{-1}\,\mathrm{Hess}_g u - 3u^{-2}du\otimes du + u^{-2}|du|_g^2 g. \end{aligned}$$

Adding these identities cancels the Hessian terms:

$$\begin{aligned} \mathrm{Ric}_{\bar{g}} + 2\,\mathrm{Hess}_{\bar{g}}\varphi + d\varphi\otimes d\varphi &= \mathrm{Ric}_g - d\varphi\otimes d\varphi - \Delta_g\varphi\, g \\ &= \mathrm{Ric}_g + \frac{f}{u}g + u^{-2}(|du|_g^2 g - du\otimes du) \\ &\geq mu^2 g = m\bar{g}. \end{aligned} \tag{A.6}$$

The last inequality uses $\mathrm{Ric}_g \geq 0$ and

$$(|du|_g^2 g - du\otimes du)(V,V) = |du|_g^2|V|_g^2 - du(V)^2 \geq 0.$$

For scalar curvature, (A.4) and (A.5) give

$$\begin{aligned} \mathrm{Scal}_{\bar{g}} &= u^{-2}(\mathrm{Scal}_g - 6\Delta_g\varphi - 6|d\varphi|_g^2) \\ &= u^{-2}\left(\mathrm{Scal}_g - 6\frac{\Delta_g u}{u} + 6\frac{|du|_g^2}{u^2} - 6\frac{|du|_g^2}{u^2}\right) \\ &= u^{-3}(-6\Delta_g u + \mathrm{Scal}_g\, u) \\ &= u^{-3}(6f + \mathrm{Scal}_g\, u) \geq 6m. \end{aligned} \tag{A.7}$$

Moreover,

$$|d\varphi|_{\bar{g}}^2 = e^{-2\varphi}|d\varphi|_g^2 = u^{-4}|du|_g^2 \leq \frac{\rho^4}{L^4}\frac{D^2}{\rho^4} = B.$$

For completeness, let $\alpha : [0,1] \to M$ be a piecewise smooth curve from p to x. Since ρ is 1-Lipschitz for d_g, $\rho\circ\alpha$ is absolutely continuous and

$$|(\rho\circ\alpha)'| \leq |\alpha'|_g \quad \text{almost everywhere.}$$

Thus

$$\mathrm{Length}_{\bar g}(\alpha) \geq L \int_0^1 \frac{|(\rho\circ\alpha)'|}{\rho\circ\alpha}\,dt \geq L\log\rho(x),$$

and hence

$$d_{\bar g}(p,x) \geq L\log\rho(x). \tag{A.8}$$

Every closed $\bar g$-ball about p is therefore a closed subset of a compact g-ball. Here we use that g and $\bar g$ induce the same topology. Hopf–Rinow proves completeness of $(M,\bar g)$ [27, Theorem 5.7.1].

Finally,

$$\varphi \leq \log U, \qquad -\varphi \geq \log\rho - \log U.$$

Thus $-\varphi$ is bounded below and has compact sublevels. □

A.3. **Proof of Theorem 3.3.** Throughout this subsection, retain the data of Proposition 3.2 and put

$$\psi = \frac{\varphi}{3}, \qquad q = u^{2/3}g = e^{2\psi}g = e^{-4\varphi/3}\bar g,$$
$$\Phi = \log U, \qquad \delta = \frac{m}{8}e^{-\sqrt{B}} > 0.$$

For this second conformal change, take $\eta = \psi = \varphi/3$ in (A.2) and (A.4). Since $d\psi = d\varphi/3$ and $\Delta_g\psi = \Delta_g\varphi/3$,

$$\begin{aligned}\mathrm{Hess}_q h &= \mathrm{Hess}_g h - \frac{1}{3}\left(d\varphi\otimes dh + dh\otimes d\varphi - \langle d\varphi, dh\rangle_g g\right),\\ \mathrm{Scal}_q &= u^{-2/3}\left(\mathrm{Scal}_g - 2\Delta_g\varphi - \frac{2}{3}|d\varphi|_g^2\right).\end{aligned} \tag{A.9}$$

Lemma A.1 (Proper ray Busemann functions)**.** *Let $\gamma:[0,\infty)\to M$ be a $\bar g$-unit-speed ray with $\gamma(0)=p$. Then*

$$b(x) = \lim_{t\to\infty}\left(t - d_{\bar g}(x,\gamma(t))\right)$$

is 1-Lipschitz for $d_{\bar g}$, bounded below, and has compact sublevels.

Proof. We first prove a uniform annular diameter bound.

The smooth four-dimensional case of [8, Theorem 4.8] gives, for all sufficiently large r, a path-connected set C_r such that

$$B^g_{4r}(p)\setminus B^g_{3r}(p) \subset C_r \subset B^g_{5r}(p)\setminus B^g_{2r}(p).$$

Write $\theta = \mathrm{AVR}(g) > 0$. If $z_1,\dots,z_N \in C_r$ are $r/4$-separated, then the balls $B^g_{r/8}(z_i)$ are disjoint and contained in $B^g_{41r/8}(p)$. Volume comparison gives

$$N\theta\omega_4(r/8)^4 \leq \mathrm{Vol}_g(B^g_{41r/8}(p)) \leq \omega_4(41r/8)^4.$$

Thus $N \leq 41^4/\theta$, independently of r.

Choose a maximal such separated set. Its $r/4$-balls cover C_r, and the intersection graph of this cover relative to C_r is connected. For $a,b\in C_r$

with $d_g(a,b) < r$, a minimizing g-segment between them stays outside $B^g_r(p)$. Hence

$$d_{\bar g}(a,b) \le \frac{U}{1+r}\, d_g(a,b) < U.$$

A simple chain in the finite covering graph uses at most $N-1$ edges. Including the two endpoint-to-center segments therefore gives

$$\operatorname{diam}_{\bar g}(C_r) \le D_0, \qquad D_0 = \big(\lceil 41^4/\theta \rceil + 2\big)U,$$

independently of r. For every sufficiently large R, take $r = R/3$. Since metric balls are open,

$$\{x : d_g(p,x) = R\} \subset B^g_{4r}(p) \setminus B^g_{3r}(p) \subset C_r.$$

Thus every such distance sphere has $\bar g$-diameter at most D_0.

The limit defining b exists because its defining terms are nondecreasing and bounded above by $d_{\bar g}(p,x)$. The triangle inequality also gives

$$|b(x) - b(y)| \le d_{\bar g}(x,y).$$

The ray γ escapes every compact set, so $d_g(p,\gamma(t)) \to \infty$. For x outside a fixed compact set, choose $s \ge 0$ such that

$$d_g(p,\gamma(s)) = d_g(p,x).$$

Then

$$d_{\bar g}(x,\gamma(s)) \le D_0, \qquad d_{\bar g}(p,x) \le s + D_0,$$

and consequently

$$b(x) \ge s - D_0 \ge d_{\bar g}(p,x) - 2D_0 \ge L \log \rho(x) - 2D_0.$$

Together with continuity on the remaining compact set, this proves boundedness below and compactness of all sublevels. □

Lemma A.2 (Weighted lower supports)**.** *Let b be as in Lemma A.1. For every compact set $K \subset M$, there is $H_K > 0$ such that each $x \in K$ admits a smooth local lower support h for b:*

$$h \le b \text{ near } x, \qquad h(x) = b(x).$$

At the contact point,

$$\begin{aligned} |dh|_{\bar g} = 1, \qquad & \operatorname{Hess}_{\bar g} h(\nabla_{\bar g} h, \nabla_{\bar g} h) = 0, \\ \Delta_{\bar g} h - 2\langle d\varphi, dh\rangle_{\bar g} \ge \delta, \\ |\operatorname{Hess}_{\bar g} h|_{\bar g} \le H_K. \end{aligned} \tag{A.10}$$

Proof. The asymptotic-ray construction [27, Proposition 7.3.8], with our Busemann sign convention, gives a $\bar g$-unit-speed ray σ from x satisfying

$$b(\sigma(s)) = b(x) + s.$$

Set

$$\varphi_\sigma(s) = \varphi(\sigma(s)), \qquad T = \max\left\{2, \frac{32 e^{\Phi - \varphi(x) + \sqrt{B}}}{m}\right\},$$

and, for $0 \le s \le T$, put

$$\ell(s) = 1 - \frac{s}{T}, \qquad \zeta(s) = e^{(\varphi_\sigma(s) - \varphi_\sigma(0))/2}\ell(s).$$

We have

$$\varphi_\sigma \le \Phi, \qquad |\varphi_\sigma'| \le \sqrt{B}, \qquad \zeta(0) = 1, \qquad \zeta(T) = 0.$$

Let $\Pi_s : T_xM \to T_{\sigma(s)}M$ be parallel transport for $\bar{g}$. For small $v \in T_xM$, define

$$c_v(s) = \exp^{\bar{g}}_{\sigma(s)}(\zeta(s)\Pi_s v),$$
$$h(\exp^{\bar{g}}_x v) = b(x) + T - \mathrm{Length}_{\bar{g}}(c_v).$$

For sufficiently small v, the curves c_v have nonzero velocity, so their lengths depend smoothly on v. Their initial and final points are $\exp^{\bar{g}}_x v$ and $\sigma(T)$. Since $b(\sigma(T)) = b(x) + T$ and b is 1-Lipschitz,

$$h(\exp^{\bar{g}}_x v) \le b(\sigma(T)) - d_{\bar{g}}(\exp^{\bar{g}}_x v, \sigma(T)) \le b(\exp^{\bar{g}}_x v).$$

Equality holds at $v = 0$.

Choose parallel $\bar{g}$-orthonormal fields E_1, E_2, E_3 perpendicular to $\dot{\sigma}$ and put

$$I_i = \int_0^T \left(\zeta'^2 - \zeta^2\, \bar{g}(R^{\bar{g}}(E_i, \dot{\sigma})\dot{\sigma}, E_i)\right) ds.$$

The first and second variation formulas [27, Section 5.4 and Theorem 6.1.4] give, at x,

$$\nabla_{\bar{g}} h = \dot{\sigma}(0),$$
$$\mathrm{Hess}_{\bar{g}}\, h(\dot{\sigma}(0), \dot{\sigma}(0)) = 0,$$
$$\Delta_{\bar{g}} h = -\sum_{i=1}^{3} I_i.$$

For clarity, the radial Hessian identity follows directly from the tangential variation. Extending σ slightly backwards, we have

$$c_{\tau\dot{\sigma}(0)}(s) = \sigma(s + \tau\zeta(s)).$$

For sufficiently small $|\tau|$, $1 + \tau\zeta'(s) > 0$, so

$$\mathrm{Length}_{\bar{g}}(c_{\tau\dot{\sigma}(0)}) = \int_0^T (1 + \tau\zeta'(s))\, ds = T - \tau.$$

Hence $h(\exp^{\bar{g}}_x(\tau\dot{\sigma}(0))) = b(x) + \tau$.

Along σ, (A.6) gives

$$\mathrm{Ric}_{\bar{g}}(\dot{\sigma}, \dot{\sigma}) \ge m - 2\varphi_\sigma'' - (\varphi_\sigma')^2.$$

Integration by parts therefore yields

$$\sum_{i=1}^{3} I_i + 2\varphi_\sigma'(0) \le \int_0^T \left(3\zeta'^2 - 4\varphi_\sigma'\zeta\zeta' + (\varphi_\sigma')^2\zeta^2 - m\zeta^2\right) ds$$
$$= \int_0^T e^{\varphi_\sigma - \varphi_\sigma(0)} \left(4\ell'^2 - \left(\ell' + \frac{\varphi_\sigma'\ell}{2}\right)^2 - m\ell^2\right) ds.$$

Since $T \geq 2$, on $[0,1]$ we have

$$\ell \geq \frac{1}{2}, \qquad e^{\varphi_\sigma - \varphi_\sigma(0)} \geq e^{-\sqrt{B}}.$$

Consequently,

$$\begin{aligned}\sum_{i=1}^{3} I_i + 2\varphi_\sigma'(0) &\leq \frac{4e^{\Phi-\varphi_\sigma(0)}}{T} - \frac{m}{4}e^{-\sqrt{B}} \\ &\leq -\frac{m}{8}e^{-\sqrt{B}} = -\delta.\end{aligned}$$

Thus

$$\Delta_{\bar{g}} h - 2\langle d\varphi, dh\rangle_{\bar{g}} = -\sum_{i=1}^{3} I_i - 2\varphi_\sigma'(0) \geq \delta.$$

For $x \in K$, the chosen lengths T are uniformly bounded. All corresponding coray segments lie in a compact $\bar{g}$-neighborhood of K. On this neighborhood the curvature is bounded, while ζ and ζ' are uniformly bounded. More explicitly, let T_K bound T, let Z_0, Z_1 bound $|\zeta|, |\zeta'|$, and let K_0 bound the absolute sectional curvature on this compact neighborhood. For $v \in T_x M$, write $v^\perp = v - \bar{g}(v, \dot\sigma(0))\dot\sigma(0)$. The length second variation depends only on this perpendicular part and gives

$$\begin{aligned}|\mathrm{Hess}_{\bar{g}}\, h(v,v)| &\leq \int_0^T (\zeta'^2 + K_0\zeta^2)|v^\perp|_{\bar{g}}^2\, ds \\ &\leq T_K(Z_1^2 + K_0 Z_0^2)|v|_{\bar{g}}^2.\end{aligned}$$

In dimension four the tensor norm is at most twice the operator norm, so $H_K = 2T_K(Z_1^2 + K_0 Z_0^2) + 1$ is an admissible bound in (A.10). □

Lemma A.3 (Increasing reparametrization)**.** *There is a smooth strictly increasing function $\chi : \mathbb{R} \to \mathbb{R}$ such that $\chi \circ b$ is proper, bounded below, and belongs to Shen's class $C(3)$ for q.*

Proof. At each x, fix a calibrated coray and the lower support h_x constructed in Lemma A.2, and retain these choices throughout. Apply (A.2) with base metric $\bar{g}$ and conformal exponent $-2\varphi/3$, since $q = e^{-4\varphi/3}\bar{g}$. This gives

$$\mathrm{Hess}_q\, h = \mathrm{Hess}_{\bar{g}}\, h + \frac{2}{3}\left(d\varphi \otimes dh + dh \otimes d\varphi - \langle d\varphi, dh\rangle_{\bar{g}}\bar{g}\right).$$

At a contact point, choose a $\bar{g}$-orthonormal frame e_0, e_1, e_2, e_3 with $e_0 = \nabla_{\bar{g}} h$, and write

$$A_{ij} = \mathrm{Hess}_q\, h(e_i, e_j).$$

By (A.10),

$$A_{11} + A_{22} + A_{33} = \Delta_{\bar{g}} h - 2\langle d\varphi, dh\rangle_{\bar{g}} \geq \delta. \tag{A.11}$$

Choose a lower bound b_* for b and set

$$K_n = \{x : b(x) \leq b_* + n + 2\}, \qquad n = 0, 1, \ldots.$$

These sets are compact. Lemma A.2 and the gradient bound for φ give constants $C_n \geq 1$ such that

$$|A_{ij}| \leq C_n \qquad (x \in K_n).$$

Enlarge them so that C_n is nondecreasing.

We use the following elementary matrix estimate. If A is symmetric, $|A_{ij}| \leq C_n$, and $A_{11} + A_{22} + A_{33} \geq \delta$, then

$$\lambda \geq 4C_n + \frac{6C_n^2}{\delta} \quad \Longrightarrow \quad \operatorname{tr}_P^{\bar{g}}(\operatorname{Hess}_q h + \lambda\, dh \otimes dh) \geq \frac{\delta}{2} \tag{A.12}$$

for every three-plane $P \subset T_xM$. Indeed, put $S = A_{11} + A_{22} + A_{33}$ and let $\nu = (\nu_0, \nu')$ be a $\bar{g}$-unit normal to P. Cauchy–Schwarz and Young's inequality give

$$\sum_{i,j=1}^{3} \nu_i A_{ij} \nu_j \leq 3C_n |\nu'|^2,$$

$$2\left|\nu_0 \sum_{i=1}^{3} A_{0i}\nu_i\right| \leq \frac{\delta}{2}\nu_0^2 + \frac{6C_n^2}{\delta}|\nu'|^2.$$

Since $A_{00} \geq -C_n$,

$$\begin{aligned}
&\operatorname{tr}_P^{\bar{g}}(\operatorname{Hess}_q h + \lambda\, dh \otimes dh) \\
&\quad = S\nu_0^2 + (S + A_{00} + \lambda)|\nu'|^2 - \sum_{i,j=1}^{3} \nu_i A_{ij}\nu_j - 2\nu_0 \sum_{i=1}^{3} A_{0i}\nu_i \\
&\quad \geq \frac{\delta}{2}\nu_0^2 + \left(\delta + \lambda - 4C_n - \frac{6C_n^2}{\delta}\right)|\nu'|^2 \\
&\quad \geq \frac{\delta}{2}.
\end{aligned}$$

Choose a smooth positive function $\Lambda : \mathbb{R} \to (0, \infty)$ such that, for every $n = 0, 1, \ldots$,

$$\Lambda(t) \geq 4C_n + \frac{6C_n^2}{\delta} \qquad \text{for } b_* + n \leq t \leq b_* + n + 1.$$

Such a function exists by taking a smooth majorant of these locally bounded stripwise constants. Define

$$\chi'(t) = \exp\left(\int_{b_*}^{t} \Lambda(s)\, ds\right),$$

$$\chi(t) = b_* + \int_{b_*}^{t} \chi'(s)\, ds.$$

Then $\chi' > 0$ and $\chi''/\chi' = \Lambda$ on $\mathbb{R}$. For every $t \geq b_*$, we also have

$$\chi'(t) \geq 1, \qquad \chi(t) \geq t.$$

In particular, $\chi \circ b$ is bounded below and has compact sublevels.

Since χ is increasing, $\chi \circ h$ is a smooth local lower support for $\chi \circ b$. At its contact point,

$$\operatorname{Hess}_q(\chi \circ h) = \chi'(b(x)) \left(\operatorname{Hess}_q h + \frac{\chi''(b(x))}{\chi'(b(x))}\, dh \otimes dh \right).$$

The q-orthonormal frame corresponding to e_i is $e^{2\varphi(x)/3} e_i$. Thus (A.12) gives

$$\operatorname{tr}_P^q \operatorname{Hess}_q(\chi \circ h) \geq \frac{\delta}{2} e^{4\varphi(x)/3} \chi'(b(x)) > 0 \qquad (\dim P = 3).$$

Fix $x_0 \in M$ and choose an open neighborhood W with $\overline{W} \subset \operatorname{int} K_N$ for some N. At each $x \in W$, choose the lower support h_x above and put $S_x = \operatorname{Hess}_q(\chi \circ h_x)_x$. The three-plane estimate and the coefficient bound give

$$\operatorname{tr}_P^q S_x \geq \eta_0, \qquad |S_x(v,v)| \leq A_0 |v|_q^2,$$

for every $x \in W$, every three-plane $P \subset T_x M$, and every $v \in T_x M$. The constants are uniform on W; one may take

$$\eta_0 = \min_{K_N} \frac{\delta}{2} e^{4\varphi/3} \chi'(b) > 0,$$
$$A_0 = 1 + \max_{K_N} e^{4\varphi/3} \chi'(b) \bigl(4C_N + \Lambda(b)\bigr).$$

Indeed, $|A_{ij}| \leq C_N$ implies $|\operatorname{Hess}_q h_x(v,v)| \leq 4C_N e^{4\varphi(x)/3} |v|_q^2$, and $|dh_x|_q^2 = e^{4\varphi(x)/3}$.

We spell out the perturbation argument of [30, Section 3.1, Lemma 6]. Choose

$$0 < \varepsilon \leq \min\left\{ \frac{1}{6}, \frac{\eta_0}{18 A_0} \right\}.$$

For any triple $v_1, v_2, v_3 \in T_x M$ with $|q(v_i, v_j) - \delta_{ij}| < \varepsilon$, its Gram matrix G satisfies $\|G - I\|_{\mathrm{op}} \leq 3\varepsilon \leq 1/2$. The vectors

$$w_i = \sum_{j=1}^{3} v_j (G^{-1/2})_{ji}$$

are q-orthonormal. Setting $T = (S_x(w_i, w_j))$, we obtain

$$\begin{aligned} \sum_{i=1}^{3} S_x(v_i, v_i) &= \operatorname{tr}(TG) \\ &\geq \operatorname{tr} T - 3\|T\|_{\mathrm{op}} \|G - I\|_{\mathrm{op}} \\ &\geq \eta_0 - 9 A_0 \varepsilon \geq \frac{\eta_0}{2}. \end{aligned}$$

Taylor expansion of the smooth lower support along q-geodesics gives

$$\begin{aligned} C^q_{\chi \circ b}(x; v) &:= \liminf_{t \to 0} \frac{(\chi \circ b)(\exp_x^q(tv)) + (\chi \circ b)(\exp_x^q(-tv)) - 2(\chi \circ b)(x)}{t^2} \\ &\geq S_x(v, v). \end{aligned}$$

Consequently, for every $x \in W$ and every such triple,

$$\sum_{i=1}^{3} C^q_{\chi\circ b}(x; v_i) \geq \frac{\eta_0}{2} > \frac{\eta_0}{4}.$$

The function b is 1-Lipschitz for $\bar{g}$. Local comparison of q and $\bar{g}$ and smoothness of χ therefore make $\chi \circ b$ locally Lipschitz for q. These are the locally uniform conditions defining $C(3)$ in [30, Section 3.1]. Hence $\chi \circ b \in C(3)$ on M. □

Proof of Theorem 3.3. Use the same potential u as in Proposition 3.2. Since $u \leq U$,

$$q = u^{-4/3}\bar{g} \geq U^{-4/3}\bar{g}.$$

Consequently,

$$d_q(p, x) \geq U^{-2/3} d_{\bar{g}}(p, x).$$

A closed q-ball is contained in a compact closed $\bar{g}$-ball and is closed in their common manifold topology. It is therefore compact, and Hopf–Rinow gives completeness of q.

Since $(M, \bar{g})$ is complete and noncompact, choose a $\bar{g}$-unit-speed ray γ issuing from p, and let b be its Busemann function.

By Lemma A.1 and Lemma A.3, the function $\chi \circ b$ is proper, bounded below, and belongs to $C(3)$ for q. Apply [30, Section 3.1, Lemma 5] with approximation error equal to 1. We obtain a smooth proper Morse function F such that

$$|F - \chi \circ b| < 1, \qquad F \in C(3).$$

The error bound makes F bounded below, so F is an exhaustion. For a smooth function, Shen's $C(3)$ condition is equivalent to

$$\lambda^q_{(3)}(\mathrm{Hess}_q F) > 0.$$

At a critical point, three negative Hessian eigenvalues would contradict this inequality. Hence

$$\mathrm{ind}_x F \leq 2 \qquad (x \in \mathrm{Crit}\, F).$$

□

Appendix B. Proofs of Section 4

B.1. **Proof of Theorem 4.1.**

Lemma B.1 (Intersection of fundamental-group images)**.** *Let $X = U \cup V$, where U and V are open subsets of X such that U, V, and $U \cap V$ are path connected. For a common basepoint $z \in U \cap V$,*

$$\begin{aligned}&\mathrm{im}\big(\pi_1(U, z) \to \pi_1(X, z)\big) \cap \mathrm{im}\big(\pi_1(V, z) \to \pi_1(X, z)\big)\\ &\quad = \mathrm{im}\big(\pi_1(U \cap V, z) \to \pi_1(X, z)\big).\end{aligned}$$

Proof. Van Kampen identifies $\pi_1(X, z)$ with the group pushout [16, Theorem 1.20]. Write

$$A = \pi_1(U, z), \qquad B = \pi_1(V, z), \qquad C = \pi_1(U \cap V, z),$$

and denote the pushout by P. Let $\overline{A}, \overline{B}, \overline{C}$ be their actual images in P, and put

$$P' = \overline{A} *_{\overline{C}} \overline{B}.$$

The inclusions of these images give a homomorphism $P' \to P$. Conversely, the maps $A \to \overline{A} \to P'$ and $B \to \overline{B} \to P'$ agree on C, so the original universal property gives $P \to P'$. The composites are the identity on their generating vertex groups, hence are inverse isomorphisms. The edge maps defining P' are injective. The desired identity therefore follows from the normal-form theorem for amalgamated free products [21, Chapter IV, Section 2]. □

Proof of Theorem 4.1. Let H be a smooth strictly three-convex Morse exhaustion for q. Three negative critical Hessian eigenvalues would contradict $\lambda^q_{(3)}(\operatorname{Hess}_q H) > 0$, so every critical index is at most two. Choose strictly increasing regular values $t_i \to \infty$ with $H(p) < t_1$, and put

$$K_i = \{H \le t_i\}, \qquad Z_i = M \setminus K_i.$$

Let D_i be the component of K_i containing p. Each K_i is a compact regular domain, possibly disconnected, and each D_i is a connected compact regular domain.

Since $t_i < t_{i+1}$,

$$D_i \subset \operatorname{int} D_{i+1}.$$

For any $x \in M$, choose a path from p to x. Its image is compact, so for sufficiently large i it lies in $\{H < t_i\}$. Hence

$$\bigcup_{i \ge 1} \operatorname{int} D_i = M.$$

The Morse handle decomposition recalled in Section 2 builds each K_i from finitely many handles of indices $0, 1, 2$. Each connected component has connected boundary: 0-handles start with S^3, 1-handles merge or preserve connected boundaries, and 2-handle surgery along a circle does not disconnect a connected three-manifold. Thus Y_{D_i} is connected.

Removing finitely many disjoint compact domains with connected boundaries leaves a path-connected complement. Indeed, choose a path between two exterior points transverse to their boundaries. It has finitely many boundary crossings; replace each excursion through a removed domain by a path in its connected boundary, pushed slightly into the exterior collar. This proves that Z_i is path connected. Filling the components of K_i other than D_i shows that O_{D_i} is path connected as well.

Choose a minimizing g-ray η from p and strictly increasing times τ_i such that

$$K_i \subset B^g_{\tau_i}(p).$$

Set $x_i = \eta(\tau_i)$. Then

$$\eta([\tau_i, \infty)) \subset Z_i \subset O_{D_i}.$$

We use these points and the corresponding ray segments for all basepoint changes below.

Reading the Morse cobordism between t_i and t_j in reverse replaces an index-λ handle by an index-$(4-\lambda)$ handle. All these dual indices are at least two, so attaching them creates no fundamental-group generators. Using boundary collars and the chosen basepoint paths, we obtain

$$\pi_1(Z_j, x_j) \twoheadrightarrow \pi_1(Z_i, x_i) \qquad (j \geq i).$$

This is the reverse-handle argument in [6, proof of Proposition 4.3].

Similarly, turning each component of K_i other than D_i upside down gives a boundary-relative handle decomposition with indices $2, 3, 4$. Filling those components therefore gives

$$\pi_1(Z_i, x_i) \twoheadrightarrow \pi_1(O_{D_i}, x_i).$$

Since $Z_j \subset O_{D_j}$, the composite surjection factors as

$$\pi_1(Z_j, x_j) \longrightarrow \pi_1(O_{D_j}, x_j) \longrightarrow \pi_1(O_{D_i}, x_i).$$

Hence

$$\pi_1(O_{D_j}, x_j) \twoheadrightarrow \pi_1(O_{D_i}, x_i) \qquad (j \geq i). \tag{B.1}$$

The smooth four-dimensional case of [8, Theorem 4.8], with the open-core construction at the beginning of the proof of [8, Proposition 6.6], gives constants $N \in \mathbb{N}$ and $R_0 > 0$ such that, for every $r > R_0$, there is an open path-connected set A_r with

$$B^g_{4r}(p) \setminus B^g_{3r}(p) \subset A_r \subset B^g_{5r}(p) \setminus B^g_{2r}(p)$$

and

$$|\mathrm{im}\,[\pi_1(A_r, z) \longrightarrow \pi_1(B^g_{6r}(p) \setminus B^g_r(p), z)]| \leq N \qquad (z \in A_r). \tag{B.2}$$

Fix i and an arbitrary finite subset $S \subset \pi_1(O_{D_i}, x_i)$. Choose based loops representing its elements, and take $r > R_0$ large enough that

$$D_i \subset B^g_r(p), \qquad \tau_i < 3r,$$

and all these loops lie in $B^g_{3r}(p)$. Set $z = \eta(7r/2)$ and define

$$\mathcal{U}_r = (A_r \cup B^g_{4r}(p)) \setminus D_i,$$
$$\mathcal{V}_r = A_r \cup (M \setminus \overline{B^g_{3r}(p)}).$$

These are open subsets of O_{D_i} satisfying

$$\mathcal{U}_r \cup \mathcal{V}_r = O_{D_i}, \qquad \mathcal{U}_r \cap \mathcal{V}_r = A_r.$$

Both sets are path connected. Indeed, every point of $\mathcal{U}_r \setminus A_r$ has g-distance less than $3r$ from p, whereas every point of $\mathcal{V}_r \setminus A_r$ has distance at least $4r$. Join such a point to z by a path in O_{D_i} and stop at its first intersection with the sphere of radius $7r/2$. In the first case the initial segment lies in

$B^g_{4r}(p) \setminus D_i$; in the second case it lies in $M \setminus \overline{B^g_{3r}(p)}$. The stopping point belongs to A_r and can be joined to z inside A_r.

Transport S to $\pi_1(O_{D_i}, z)$ along the ray segment. Its chosen representatives, including their basepoint paths, lie in $\mathcal{U}_r$.

Choose $j \geq i$ such that

$$\overline{B^g_{4r}(p)} \subset \operatorname{int} D_j.$$

Since $D_j \subset K_j \subset B^g_{\tau_j}(p)$, we have $\tau_j > 4r$. By (B.1), the same transported classes have representatives in O_{D_j}, joined to z by the ray segment from z to x_j. Since O_{D_j} lies outside $\overline{B^g_{4r}(p)}$ and this ray segment has radial coordinate at least $7r/2$, the resulting representatives lie in $\mathcal{V}_r$.

Lemma B.1 therefore places every transported element of S in

$$\operatorname{im}(\pi_1(A_r, z) \to \pi_1(O_{D_i}, z)).$$

The latter map factors through

$$A_r \subset B^g_{6r}(p) \setminus B^g_r(p) \subset O_{D_i}.$$

Thus (B.2) gives $|S| \leq N$. Since S was an arbitrary finite subset,

$$|\pi_1(O_{D_i}, x_i)| \leq N.$$

The same N works for every i, so $N_H = N$ can be chosen to depend only on (M, g), independently of H.

Finally, $Y_{D_i} \subset H^{-1}(t_i)$ is a regular level with outward normal $\nabla_q H/|dH|_q$. Hence

$$H_q(Y_{D_i}) = \frac{\operatorname{tr}^q_{(\nabla_q H)^\perp} \operatorname{Hess}_q H}{|dH|_q} \geq \frac{\lambda^q_{(3)}(\operatorname{Hess}_q H)}{|dH|_q} > 0.$$

□

Appendix C. Proofs of Section 5

Throughout this section, retain the conformal and Morse data and the compact regular domains C_i constructed above. In particular, q and $\bar{g}$ are complete, and F is a smooth strictly three-convex Morse exhaustion for q.

C.1. **Proof of Proposition 5.3.**

Proof. Choose $\sigma \in C^\infty(\mathbb{R})$ such that

$$0 \leq \sigma \leq 1, \qquad \sigma(t) = 0 \quad (t \leq 1/3), \qquad \sigma(t) = 1 \quad (t \geq 2/3),$$

and put $C_\sigma = \sup_{\mathbb{R}} |\sigma'|$. Choose

$$R > \max\left\{6, \frac{2\pi C_\sigma}{\sqrt{m}}\right\}.$$

This choice is independent of the plateau values below.

Set $A = C_i$ and

$$r_A(x) = \inf_{y \in A} d_{\bar{g}}(x, y).$$

Completeness of $\bar{g}$ and compactness of A imply that $\{r_A \le R+2\}$ is compact. Choose $j > i$ such that

$$\{r_A \le R+2\} \subset \operatorname{int} C_j,$$

and put $B = C_j$. Thus $r_A > R+2$ on Y_B.

The distance function r_A is 1-Lipschitz. By [2, Theorem 1], there is $r \in C^\infty(M)$ with

$$|r - r_A| \le 1, \qquad |dr|_{\bar{g}} \le 2.$$

Consequently,

$$r \le 1 < R/3 \quad \text{on } A, \qquad r > R+1 > 2R/3 \quad \text{on } Y_B.$$

Choose constants a_0, a_1 such that

$$a_0 > \max_{Y_A} u^{-2/3} H_q(Y_A),$$

$$0 < a_1 < \min\left\{a_0, \min_{Y_B} u^{-2/3} H_q(Y_B)\right\}.$$

The second minimum is positive by Theorem 4.1. Put

$$\beta_\ell = \arctan \frac{a_\ell}{\sqrt{m}} \quad (\ell = 0, 1),$$

$$\beta(t) = (1 - \sigma(t/R))\beta_0 + \sigma(t/R)\beta_1,$$

and define

$$a(t) = \sqrt{m}\, \tan \beta(t), \qquad h = a \circ r.$$

Since

$$0 < \beta_1 \le \beta(t) \le \beta_0 < \pi/2,$$

both functions are smooth and positive. Moreover,

$$a = a_0 \quad (t \le R/3), \qquad a = a_1 \quad (t \ge 2R/3).$$

Hence $h = a_0$ on A and $h = a_1$ on Y_B, giving

$$H_q(Y_A) < u^{2/3} h, \qquad H_q(Y_B) > u^{2/3} h.$$

Since $|\beta_0 - \beta_1| < \pi/2$,

$$|\beta'| \le \frac{\pi C_\sigma}{2R} < \frac{\sqrt{m}}{4}.$$

Thus

$$a' = \frac{a^2 + m}{\sqrt{m}}\, \beta', \qquad |a'| \le \frac{a^2 + m}{4}.$$

Using $|dr|_{\bar{g}} \le 2$, we obtain

$$2|dh|_{\bar{g}} = 2|a'(r)|\, |dr|_{\bar{g}} \le h^2 + m.$$

This proves (5.2).

To prove (5.3), put

$$z = u^{-4/3} |du|_g, \qquad k = u^{2/3} h.$$

The conformal scalar-curvature formula from Section 2, together with $-\Delta_g u = f$, gives

$$\begin{aligned}\operatorname{Scal}_q &= u^{-2/3}\operatorname{Scal}_g + 2fu^{-5/3} + \frac{4}{3}u^{-8/3}|du|_g^2 \\ &\geq 2mu^{4/3} + \frac{4}{3}z^2.\end{aligned}$$

Also,

$$\begin{aligned}|dk|_q &\leq \frac{2}{3}hu^{-2/3}|du|_g + u^{1/3}|dh|_g \\ &= \frac{2}{3}kz + u^{4/3}|dh|_{\bar{g}}.\end{aligned}$$

Therefore,

$$\begin{aligned}\operatorname{Scal}_q + \frac{4}{3}k^2 - 2|dk|_q &\geq \frac{4}{3}\left(z - \frac{k}{2}\right)^2 + u^{4/3}(2m + h^2 - 2|dh|_{\bar{g}}) \\ &\geq mu^{4/3} > 0.\end{aligned}$$

□

C.2. **Proof of Theorem 5.5.**

Proof. Fix the given orientation of M. In our applications, an orientation exists because M is contractible. Put $\mathsf{p} = (\psi/3)q$. For every hypersurface $\Sigma \subset B$,

$$\operatorname{tr}^q_{T\Sigma}\mathsf{p} = \psi > 0, \qquad |\operatorname{tr}^q_{T\Sigma}\mathsf{p}| = \psi.$$

Apply [9, Theorem 5.1] to the complete initial data set (M, q, p), with $n = 4$ and

$$\Omega = \operatorname{int} B \setminus A, \qquad \partial_1\Omega = Y_B, \qquad \partial_2\Omega = Y_A.$$

The normal pointing out of Ω along Y_B is the outward normal of B, and the normal pointing into Ω along Y_A is the outward normal of A. The hypotheses therefore read

$$H_q(Y_B) > \psi, \qquad H_q(Y_A) \leq \psi.$$

The theorem gives an open set $\Omega' \subset \Omega$ with

$$\partial\Omega' = Y_B \sqcup \Sigma',$$

where Σ' is an embedded $C^{2,\alpha}$ hypersurface satisfying $H_q = \psi$ with respect to the normal pointing into Ω'. Since $\psi > 0$, this is the smooth prescribed-mean-curvature equation. In local graph coordinates it is uniformly elliptic on smaller neighborhoods. Schauder bootstrapping [14, Theorem 6.17] therefore makes Σ' smooth.

Put

$$E = \operatorname{int} B \setminus \Omega'.$$

Near Y_B, the smooth domain Ω' occupies the inward side of Y_B. Compactness gives an inward collar of Y_B contained in Ω'. Thus E stays away from Y_B and is compact in M. Moreover,

$$A \subseteq E \subset \operatorname{int} B, \qquad Y_E = \Sigma',$$

and the outward normal of E points into Ω'. Hence E is a compact regular domain with $H_q(Y_E) = \psi$. Its boundary is nonempty because $\varnothing \neq A \subseteq E \neq M$ and M is connected.

Let D be a compact regular domain with

$$A \subseteq D \subset \operatorname{int} B, \qquad H_q(Y_D) \leq \psi.$$

Then

$$\Omega'' = \operatorname{int} B \setminus D \subset \Omega$$

has boundary $Y_B \sqcup Y_D$, and the normal pointing into Ω'' is the outward normal of D. The outermost property in [9, Theorem 5.1] gives $\Omega' \subset \Omega''$, hence $D \subseteq E$.

Finally, compactness of Y_A and the strict inequality $H_q(Y_A) < \psi$ give a sufficiently small $t > 0$ such that

$$A_t = \{x : d_q(x, A) \leq t\}$$

is a compact regular domain in $\operatorname{int} B$ with $H_q(Y_{A_t}) < \psi$. The enclosure property gives $A_t \subseteq E$, and hence $A \subset \operatorname{int} E$. □

C.3. **Proof of Proposition 5.6.**

Proof. The function $k = u^{2/3}h$ is positive, and (5.2) gives

$$H_q(Y_A) < k, \qquad H_q(Y_B) > k.$$

Theorem 5.5, applied with $\psi = k$, gives a compact regular domain T with

$$A \subset \operatorname{int} T, \qquad T \subset \operatorname{int} B, \qquad H_q(Y_T) = k,$$

containing every compact regular domain L with

$$A \subseteq L \subset \operatorname{int} B, \qquad H_q(Y_L) \leq k.$$

If moreover $T \subseteq L$, then $L \subseteq T$, so $L = T$. Thus T is weakly outermost in the sense of Definition 5.4.

We verify PSC on each connected component Σ of Y_T. Write $q_\Sigma = q|_{T\Sigma}$ and let ν be the outward q-unit normal of T. Under a normal variation with speed η,

$$H_q' = -\Delta_{q_\Sigma}\eta - \big(\mathrm{Ric}_q(\nu,\nu) + |\mathrm{II}_q|^2_{q_\Sigma}\big)\eta,$$
$$k' = \eta\,\nu k.$$

Consequently, the linearization of $H_q - k$ is

$$\mathcal{L}_\Sigma\eta = -\Delta_{q_\Sigma}\eta - \big(\mathrm{Ric}_q(\nu,\nu) + |\mathrm{II}_q|^2_{q_\Sigma} + \nu k\big)\eta.$$

Its first eigenvalue is nonnegative. Indeed, a negative first eigenvalue and its positive eigenfunction would give a small outward deformation of Σ with $H_q - k < 0$. Leaving the other boundary components fixed would produce a larger admissible compact regular domain, contradicting weak outermostness. This is the outermost-stability argument used in [12, proof of Theorem 2.1].

It follows that, for every $\eta \in C^\infty(\Sigma)$,

$$\int_\Sigma \Big(|d\eta|^2_{q_\Sigma} - \big(\mathrm{Ric}_q(\nu,\nu) + |\mathrm{II}_q|^2_{q_\Sigma} + \nu k\big)\eta^2\Big)\, d\,\mathrm{Vol}_{q_\Sigma} \geq 0.$$

The Gauss equation and $H_q(\Sigma) = k$ give

$$2\,\mathrm{Ric}_q(\nu,\nu) = \mathrm{Scal}_q - \mathrm{Scal}_{q_\Sigma} + k^2 - |\mathrm{II}_q|^2_{q_\Sigma}.$$

Using

$$|\mathrm{II}_q|^2_{q_\Sigma} \geq k^2/3, \qquad \nu k \geq -|dk|_q,$$

we obtain

$$\int_\Sigma \left(2|d\eta|^2_{q_\Sigma} + \mathrm{Scal}_{q_\Sigma}\,\eta^2\right) d\,\mathrm{Vol}_{q_\Sigma} \geq \int_\Sigma \left(\mathrm{Scal}_q + \frac{4}{3}k^2 - 2|dk|_q\right)\eta^2 d\,\mathrm{Vol}_{q_\Sigma}.$$

By compactness and (5.3),

$$\delta_\Sigma = \min_\Sigma \left(\mathrm{Scal}_q + \frac{4}{3}k^2 - 2|dk|_q\right) > 0.$$

Adding the nonnegative term $6\int_\Sigma |d\eta|^2_{q_\Sigma}\, d\,\mathrm{Vol}_{q_\Sigma}$ to the left-hand side gives

$$\int_\Sigma \left(8|d\eta|^2_{q_\Sigma} + \mathrm{Scal}_{q_\Sigma}\,\eta^2\right) d\,\mathrm{Vol}_{q_\Sigma} \geq \delta_\Sigma \int_\Sigma \eta^2\, d\,\mathrm{Vol}_{q_\Sigma}.$$

Thus the conformal Laplacian

$$-8\Delta_{q_\Sigma} + \mathrm{Scal}_{q_\Sigma}$$

has first eigenvalue $\lambda_\Sigma \geq \delta_\Sigma > 0$. Let $\zeta > 0$ be a corresponding smooth eigenfunction. The three-dimensional conformal scalar-curvature formula yields

$$\mathrm{Scal}_{\zeta^4 q_\Sigma} = \zeta^{-5}(-8\Delta_{q_\Sigma}\zeta + \mathrm{Scal}_{q_\Sigma}\,\zeta) = \lambda_\Sigma \zeta^{-4} > 0.$$

Therefore every component of Y_T admits PSC. □

C.4. **Proof of Lemma 5.8.**

Lemma C.1 (Inward collar data)**.** *Let K be a connected compact regular domain with $H_q(Y_K) > 0$, and let $A \subset \operatorname{int} K$ be compact. There are a connected compact regular domain Γ, $r \in C^\infty(M)$, and $c < 0$ such that*

$$K = \{r \leq 0\}, \qquad \Gamma = \{r \leq c\},$$
$$A \subset \operatorname{int}\Gamma, \qquad \Gamma \subset \operatorname{int} K.$$

The pairs (Γ, Y_Γ) and (K, Y_K) are diffeomorphic, and on $K \setminus \operatorname{int}\Gamma$,

$$dr \neq 0, \qquad \mathrm{tr}^q_{(\nabla_q r)^\perp} \mathrm{Hess}_q\, r > 0. \tag{C.1}$$

Proof. Choose a tubular neighborhood of Y_K with signed q-distance coordinate s, negative inside K. Compactness and $H_q(Y_K) > 0$ allow the collar to be chosen so that its parallel levels are strictly mean convex [18, 27]. Shrink it so that it is disjoint from A.

Extend the signed coordinate to a smooth function r, constant and negative in the deeper interior of K and constant and positive away from the exterior collar, such that $K = \{r \leq 0\}$. Choose $c < 0$ sufficiently close to zero that

$$\Gamma = \{r \leq c\}$$

contains A in its interior and $r = s$ throughout $\{c \le r \le 0\}$. Then $\Gamma \subset \operatorname{int} K$, and on the closed collar $K \setminus \operatorname{int} \Gamma$,

$$dr \neq 0, \qquad \operatorname{tr}^q_{(\nabla_q r)^\perp} \operatorname{Hess}_q r > 0.$$

In particular, Y_Γ is smooth and strictly mean convex.

An increasing change of the collar coordinate, equal to the identity near its ends and sending c to 0, extends by the identity to an ambient diffeomorphism taking (Γ, Y_Γ) to (K, Y_K). This proves connectedness and the asserted diffeomorphism. □

Proof of Lemma 5.8. Apply Lemma C.1 to the component K in Corollary 5.7. The boundary diffeomorphism transfers a PSC metric from Y_K to Y_Γ. Equation (C.1) gives

$$H_q(Y_\Gamma) = \frac{\operatorname{tr}^q_{(\nabla_q r)^\perp} \operatorname{Hess}_q r}{|dr|_q} > 0.$$

Corollary 5.2 then gives

$$\pi_1(Y_\Gamma) \twoheadrightarrow \pi_1(\Gamma).$$

□

C.5. **Uniform protection for the relative maximum principle.**

Remark (A quantitative rank-one correction). Let S be a symmetric form on a Riemannian four-space, and let $\xi \neq 0$ be a covector. Suppose

$$\operatorname{tr}^q_{\ker \xi} S \ge \delta > 0, \qquad \|S\|_{\mathrm{op},q} \le C, \qquad |\xi|_q \ge \alpha > 0.$$

Put $n = \xi^\sharp / |\xi|_q$. For a unit normal $z = an + v$ to a three-plane P, with $v \perp n$ and $\tau = |v|_q^2 = 1 - a^2$, we have

$$\begin{aligned} \operatorname{tr}^q_P(S + L\xi \otimes \xi) &= \operatorname{tr}^q_{n^\perp} S + S(n,n) - S(z,z) + L|\xi|_q^2 \tau \\ &\ge \delta - 2C\sqrt{\tau} - 2C\tau + L\alpha^2\tau \\ &\ge \frac{\delta}{2} + \left(L\alpha^2 - 2C - \frac{2C^2}{\delta} \right) \tau. \end{aligned}$$

The last step uses $2C\sqrt{\tau} \le \delta/2 + 2C^2\tau/\delta$. Consequently, for every three-plane P,

$$\begin{aligned} & L\alpha^2 \ge 2C + \frac{2C^2}{\delta} \\ \implies \quad & \operatorname{tr}^q_P(S + L\xi \otimes \xi) \ge \frac{\delta}{2}. \end{aligned} \tag{C.2}$$

For continuous data on a compact set with strict tangential trace and nonvanishing ξ, the constants can be chosen uniformly.

Lemma C.2 (Uniform protection from a mean-convex boundary)**.** *Let (M, q) be complete and connected, let B be compact, and suppose*

$$\Gamma \subset \operatorname{int} T, \qquad T \subseteq B,$$

where Γ is compact. Given $\delta > 0$, there is $a > 0$, chosen independently of D and S, with the following property. Let D be a compact regular domain satisfying

$$T \subseteq D \subseteq B, \qquad H_q(Y_D) \geq \delta,$$

and let $S \subseteq D$ be compact with

$$\mathsf{MP}_q(S; M \setminus \Gamma).$$

Then

$$d_q(S, Y_D) \geq 4a.$$

Proof. Choose $r_0 > 0$ with

$$d_q(\Gamma, M \setminus \operatorname{int} T) \geq r_0,$$

and choose $\Lambda \geq 0$ such that $\mathrm{Ric}_q \geq -\Lambda q$ on B. Set

$$a = \frac{1}{4} \min\left\{ r_0, \frac{\delta}{\Lambda + 1} \right\} > 0.$$

The assertion is immediate if $S = \varnothing$.

A point of $S \cap Y_D$ would lie outside Γ. A local defining function for D, followed by a sufficiently convex increasing reparametrization, would have positive Hessian trace on every three-plane and a local maximum on S there. This follows from (C.2) and $H_q(Y_D) > 0$, and contradicts the relative maximum principle. Thus $S \subset \operatorname{int} D$.

Realize

$$l = d_q(S, Y_D) > 0$$

at $x \in S$ and $y \in Y_D$. If $x \in \Gamma$, then $l \geq r_0$. Suppose henceforth that $x \notin \Gamma$.

Let $\gamma : [0, l] \to M$ be a minimizing geodesic from y to x. It starts in the inward normal direction and lies in D: any later intersection with Y_D would give a boundary point closer to x. We construct a smooth upper support for the boundary distance without assuming that this distance is smooth at x.

Let $\Pi_s : T_y M \to T_{\gamma(s)} M$ be parallel transport. For small $v \in T_x M$, write

$$v = \Pi_l(w + b\dot{\gamma}(0)), \qquad w \perp \dot{\gamma}(0),$$

and define

$$a_0(w) = (\exp_y^q)^{-1}(\exp_y^{Y_D}(w)),$$

where $\exp_y^{Y_D}$ is the intrinsic exponential map of Y_D. Set

$$c_v(s) = \exp^q_{\gamma(s)} \left(\Pi_s \left[\left(1 - \frac{s}{l} \right) a_0(w) + \frac{s}{l} (w + b\dot{\gamma}(0)) \right] \right).$$

Then $c_v(0) \in Y_D$ and $c_v(l) = \exp_x^q(v)$. For sufficiently small v, the curves have nonzero velocity, so

$$\ell(\exp_x^q(v)) = \mathrm{Length}_q(c_v)$$

defines a smooth function near x. It satisfies

$$d_q(\,\cdot\,, Y_D) \le \ell, \qquad \ell(x) = l, \qquad \nabla_q \ell(x) = \dot\gamma(l).$$

For variations with $b = 0$, the first-variation fields along γ are the parallel fields $\Pi_s w$. The initial endpoint curves are intrinsic geodesics of Y_D, and the final endpoint curves are ambient geodesics. The first and second variation formulas [27] therefore give

$$\operatorname{tr}^q_{\dot\gamma(l)^\perp} \operatorname{Hess}_q \ell(x) = -H_q(Y_D)(y) - \int_0^l \operatorname{Ric}_q(\dot\gamma, \dot\gamma)\, ds \le -\delta + \Lambda l.$$

The boundary term is $-H_q$ because $\dot\gamma(0)$ is the inward normal.

On S near x,

$$\ell \ge d_q(\,\cdot\,, Y_D) \ge l.$$

Hence $e^{-L\ell}$ has a local maximum on S at x. If $\delta > \Lambda l$, then

$$\operatorname{Hess}_q(e^{-L\ell}) = L e^{-L\ell}(-\operatorname{Hess}_q \ell + L\, d\ell \otimes d\ell)$$

has positive trace on every three-plane for sufficiently large L. Indeed, apply (C.2) with $S = -\operatorname{Hess}_q \ell$, $\xi = d\ell$, and tangential trace lower bound $\delta - \Lambda l > 0$. This contradicts the relative maximum principle at x.

Thus, when $x \notin \Gamma$, one has $\delta \le \Lambda l$. Together with the case $x \in \Gamma$, this gives

$$l \ge \min\left\{r_0, \frac{\delta}{\Lambda + 1}\right\} = 4a.$$

□

C.6. **Proof of Lemma 5.10.**

Proof. Assume $S \neq \varnothing$. Put $\delta = \min_B k > 0$ and choose the protection radius a from Lemma C.2 before choosing any inflation or forcing bump. Since $H_q(Y_B) > k \ge \delta$, applying that lemma to $D = B$ gives

$$d_q(S, Y_B) \ge 4a.$$

Choose $0 \le \beta \le 1$ smooth, equal to one near S, with compact support in

$$\{x : d_q(x, S) < a/2\}.$$

Let b_T be a smooth global defining function for T, with

$$T = \{b_T \le 0\}, \qquad Y_T = \{b_T = 0\},$$

and $db_T \neq 0$ on Y_T. Choose a regular value

$$t > \max\{0, \max_S b_T\}$$

of b_T/β on $\{\beta > 0\}$, and set

$$D = \{b_T - t\beta \le 0\}.$$

Sard's theorem applies to the smooth quotient on its open domain. On its zero level with $\beta > 0$,

$$d(b_T - t\beta) = \beta\, d(b_T/\beta) \neq 0.$$

At a zero with $\beta = 0$, nonnegativity gives $d\beta = 0$, and hence

$$d(b_T - t\beta) = db_T \neq 0.$$

Thus D is regular. Moreover, $D \subset T \cup \operatorname{supp}\beta$, so it is compact, and

$$T \subseteq D, \qquad S \subset \operatorname{int} D, \qquad D \subset \operatorname{int} B.$$

Outside $\operatorname{supp}\beta$, the domains D and T agree locally.

Choose $0 \leq \xi \leq 1$ smooth, equal to one near $\operatorname{supp}\beta$, with compact support in $\{x : d_q(x, S) < a\}$. Take

$$C > \max\{0, \max_{Y_D}(H_q(Y_D) - k)\},$$

and set $\psi = k + C\xi$. On the unchanged boundary, $H_q(Y_D) = k$; on the changed part, $\xi = 1$. Therefore,

$$H_q(Y_D) \leq \psi, \qquad \psi \geq k, \qquad \psi = k \text{ near } Y_B.$$

The original barriers remain valid:

$$H_q(Y_A) < k \leq \psi, \qquad H_q(Y_B) > k = \psi.$$

Theorem 5.5, applied with this ψ, supplies a compact regular domain E such that

$$D \subseteq E \subset \operatorname{int} B, \qquad H_q(Y_E) = \psi \geq \delta.$$

In particular, $T \subseteq E$ and $S \subset \operatorname{int} E$. Applying Lemma C.2 to E with the same radius a gives

$$d_q(S, Y_E) \geq 4a.$$

Since $\operatorname{supp}\xi \subset \{x : d_q(x, S) < a\}$, we have $\xi = 0$ on Y_E. Thus

$$H_q(Y_E) = k.$$

The weak outermost property of T forces $E = T$, and consequently $S \subset \operatorname{int} T$.

The component K is open and closed in the compact regular domain T. If $S \setminus K$ were nonempty, a maximum of F on this compact set would be a local maximum on S outside Γ, contradicting strict three-convexity. Thus $S \subset K$.

If $S \not\subset \Gamma$, maximize r on S. At a maximizing point, $r > c$, and

$$\operatorname{Hess}_q(e^{Lr}) = Le^{Lr}(\operatorname{Hess}_q r + L\, dr \otimes dr)$$

has positive trace on every three-plane for sufficiently large L, by (C.2). This again contradicts the relative maximum principle. Hence $S \subset \Gamma$. □

C.7. **The weak hull and a global defining level.** Choose a smooth convex function $\chi_a \geq |\cdot|$ equal to $|\cdot|$ outside $(-a, a)$, and set

$$\max_a(s, t) = \frac{s + t + \chi_a(s - t)}{2}.$$

Convexity gives $\chi_a(z) \leq a$ for $|z| \leq a$, and hence

$$\begin{aligned} 0 \leq \max_a(s, t) - \max(s, t) &\leq \frac{a}{2}, \\ |s - t| \geq a \quad &\Longrightarrow \quad \max_a(s, t) = \max(s, t). \end{aligned} \tag{C.3}$$

For smooth functions v, w,

$$\begin{aligned}&\operatorname{Hess}_q \max_a(v,w)\\ &\quad= t\operatorname{Hess}_q v + (1-t)\operatorname{Hess}_q w + \frac{\chi_a''(v-w)}{2}\, d(v-w)\otimes d(v-w),\end{aligned}\tag{C.4}$$

where $t = (1+\chi_a'(v-w))/2 \in [0,1]$. Thus smooth maxima preserve weak three-convexity, and preserve strict three-convexity when both inputs are strict.

Lemma C.3 (Maximum principle for the weak hull)**.** *For every nonempty compact* Γ*,*

$$\mathsf{MP}_q(\widehat{\Gamma}_3^q; M\setminus\Gamma)$$

holds. If F *is a strictly three-convex exhaustion, the hull is compact.*

Proof. The hull is closed and contains Γ. If a strictly three-convex exhaustion F exists, the hull is contained in

$$\{F \le \max_\Gamma F\},$$

and is therefore compact.

To prove the maximum principle, suppose that a smooth test function ψ has a local maximum on $\widehat{\Gamma}_3^q$ at

$$x \in \widehat{\Gamma}_3^q \setminus \Gamma$$

and has positive Hessian trace on every three-plane at x. Compactness of the Grassmannian and continuity of the Hessian give a neighborhood with a uniform positive trace margin. Subtracting a sufficiently small multiple of a local squared distance, and then a constant, gives a smooth function ψ_0 on a relatively compact neighborhood $\Omega \Subset M\setminus\Gamma$ such that

$$\begin{aligned}\psi_0(x) &= 0,\\ \lambda^q_{(3)}(\operatorname{Hess}_q \psi_0) &\ge 0 \quad \text{on } \Omega,\\ \psi_0 &\le -4d \quad \text{on } \widehat{\Gamma}_3^q \cap \partial\Omega\end{aligned}$$

for some $d > 0$.

Every compact set outside the hull admits a smooth weakly three-convex separator u with $u \le -1$ on Γ and $u > 1$ on that compact set. Indeed, choose finitely many separating functions $u_1, \dots, u_N$ such that

$$u_j \le -2 \text{ on } \Gamma$$

and the sets $\{u_j > 2\}$ cover the given compact set. Iterating smooth maxima with $(N-1)a/2 < 1$ gives the asserted separator by (C.3) and (C.4). For the empty set, use $u = -1$.

Apply this construction to the compact set

$$E = \{y \in \partial\Omega : \psi_0(y) \ge -2d\},$$

which is disjoint from the hull. Choose $A > 0$ sufficiently large and put

$$v = \max_{d/4}(-d, Au - 3d).$$

Since x belongs to the hull,

$$u(x) \le \max_{\Gamma} u \le -1.$$

Thus $v = -d$ on Γ and at x. On E, choose A large enough that $\psi_0 < v - d/4$. On $\partial\Omega \setminus E$, this inequality follows from $\psi_0 < -2d$ and $v \ge -d$. Hence

$$\psi_0 < v - d/4 \quad \text{on } \partial\Omega.$$

Define a global function by taking $\max_{d/4}(v, \psi_0)$ on Ω and v outside. The formulas agree near $\partial\Omega$, so the function is smooth and weakly three-convex. It equals $-d$ on Γ and 0 at x, contradicting the definition of the hull. □

Lemma C.4 (Hull equality from compact capture)**.** *In Lemma 5.10, suppose in addition that*

$$F \le b \text{ on } \Gamma, \qquad F > b \text{ on } \partial B$$

for some b. Then $\widehat{\Gamma}^q_3 = \Gamma$.

Proof. Put $S = \widehat{\Gamma}^q_3$. Lemma C.3 gives its compactness and relative maximum principle. The defining inequality for F gives $F \le b$ on S, so $S \cap \partial B = \varnothing$. Consequently,

$$S \setminus B = S \cap (M \setminus \operatorname{int} B)$$

is compact. If this set were nonempty, a maximum of F on it would be a local maximum on S outside Γ, contradicting the relative maximum principle. Thus $S \subset B$. Lemma 5.10 now gives $S \subset \Gamma$. The reverse inclusion follows from the definition. □

Lemma C.5 (A strict global defining level)**.** *Let Γ be a nonempty smooth compact regular domain with $H_q(\partial\Gamma) > 0$, and suppose*

$$\widehat{\Gamma}^q_3 = \Gamma.$$

If M has a smooth strictly three-convex exhaustion F, then there is a smooth strictly three-convex exhaustion G_0 satisfying

$$\Gamma = \{G_0 \le 0\}, \qquad \partial\Gamma = \{G_0 = 0\},$$

with $dG_0 \ne 0$ on $\partial\Gamma$.

Proof. Choose a smooth global defining function r with

$$\Gamma = \{r \le 0\}, \qquad \partial\Gamma = \{r = 0\},$$

with $dr \ne 0$ on $\partial\Gamma$ and $r = 1$ outside a compact set. Positive mean curvature gives

$$\operatorname{tr}^q_{(\nabla_q r)^\perp} \operatorname{Hess}_q r = |dr|_q H_q(\partial\Gamma) > 0.$$

Apply (C.2) uniformly on the compact boundary. By continuity, there are $L > 0$ and $0 < \epsilon < 1$ such that

$$\lambda^q_{(3)}\big(\operatorname{Hess}_q r + L\, dr \otimes dr\big) > 0 \quad \text{on } U = \{|r| < \epsilon\}.$$

Here ϵ is chosen small enough that U lies in the boundary neighborhood where this inequality holds.

Set

$$h = A(e^{Lr} - 1), \qquad A = \frac{4}{1 - e^{-L\epsilon}}.$$

Then h is strictly three-convex on U, has the sign of r, and

$$h|_{r=-\epsilon} = -4, \qquad h|_{r=0} = 0, \qquad h|_{r=\epsilon} = 4e^{L\epsilon}.$$

Put

$$C = \{r \le \epsilon/2\}, \qquad H = 4e^{L\epsilon} + 3.$$

We construct a strictly three-convex exhaustion v satisfying

$$-3 < v \le -2 \text{ on } \Gamma, \qquad v > H \text{ on } M \setminus \operatorname{int} C.$$

Choose a positive affine rescaling f of F with

$$f > -3 \text{ on } M, \qquad f \le -2 \text{ on } \Gamma.$$

Choose a sufficiently large compact sublevel P of F, containing C, such that $f > H + 4$ on $M \setminus P$. The set

$$E = P \setminus \operatorname{int} C$$

is compact and disjoint from Γ.

Hull equality and the finite separator construction in Lemma C.3 give a smooth weakly three-convex function w with

$$w \le -2 \text{ on } \Gamma, \qquad w > 2 \text{ on } E.$$

Adding a sufficiently small positive multiple of F gives a strictly three-convex function u satisfying

$$u \le -1 \text{ on } \Gamma, \qquad u > 1 \text{ on } E.$$

The inequalities are preserved because $\Gamma \cup E$ is compact.

Choose $0 < a_0 < 1$ and set

$$v = \max\nolimits_{a_0}\big(f, (H + 10)u - 6\big).$$

On Γ,

$$(H + 10)u - 6 \le -H - 16 < f - 1,$$

so $v = f$ there. Hence $-3 < v \le -2$ on Γ. On E, the second input is greater than $H + 4$; on $M \setminus P$, the first input is greater than $H + 4$. Thus $v > H$ on $M \setminus \operatorname{int} C$. Both inputs are strictly three-convex, and $v \ge f$, so v is a strictly three-convex exhaustion.

For $a = 1/10$, define

$$G_0 = \begin{cases} \max_a(v, h), & |r| < \epsilon, \\ v, & |r| \ge \epsilon. \end{cases}$$

On $r = -\epsilon$, one has $v - h > 1$. On $r = \epsilon$, one has $v - h > 3$. Thus the two formulas agree on neighborhoods of the seams.

On $\partial\Gamma$, $h = 0$ and $v \le -2$, so $G_0 = h$ in a smaller collar. For $-\epsilon < r < 0$, if $h \le -1$, then

$$G_0 \le \max(v, h) + \frac{1}{20} \le -1 + \frac{1}{20} < 0.$$

If $-1 < h < 0$, then $h - v > 1 > a$, so $G_0 = h < 0$. For $0 < r < \epsilon$, one has $G_0 \geq h > 0$. Outside U, the inner region has $v \leq -2$, while the outer region lies outside C, where $v > H > 0$. This proves

$$\Gamma = \{G_0 \leq 0\}, \qquad \partial\Gamma = \{G_0 = 0\}.$$

Since $G_0 = h$ near the boundary, $dG_0 \neq 0$ there.

Equation (C.4) gives strict three-convexity, and $G_0 \geq v$ gives boundedness below and compact sublevels. □

Lemma C.6 (Morse approximation retaining the defining level)**.** *The function G_0 in Lemma C.5 can be replaced by a smooth strictly three-convex Morse exhaustion G with the same zero sublevel and the same germ near $\partial\Gamma$. Every critical point of G has index at most two.*

Proof. Choose $0 \leq \beta \leq 1$ smooth, equal to 1 near $\{G_0 = 0\}$, with compact support contained in $\{dG_0 \neq 0\}$. On that support,

$$|dG_0|_q^2 \geq m_0 > 0.$$

Properness gives a positive lower bound for $|G_0|$ outside $\operatorname{int}\{\beta = 1\}$.

Morse functions are dense in the strong C^2 topology [17, Chapter 6, Section 1]. Choose a smooth Morse function H with sufficiently fine value, differential, and Hessian approximation to G_0. Put $E = H - G_0$ and

$$G = G_0 + (1 - \beta)E.$$

Then $G = G_0$ near the zero level and $G = H$ outside $\operatorname{supp}\beta$. The identity

$$dG(\nabla_q G_0) = |dG_0|_q^2 + (1 - \beta)dE(\nabla_q G_0) - E\, d\beta(\nabla_q G_0)$$

keeps this expression positive on $\operatorname{supp}\beta$ when the first jets are sufficiently close. Thus every critical point lies where G agrees locally with the Morse function H.

Choose a continuous function $\tau > 0$ such that

$$\operatorname{tr}_P^q \operatorname{Hess}_q G_0 \geq \tau(x)$$

for every three-plane at x. Let A_1, A_2 bound $|d\beta|_q$ and $\|\operatorname{Hess}_q \beta\|_{\mathrm{op},q}$. If

$$\max\{|E|,\ |dE|_q,\ \|\operatorname{Hess}_q E\|_{\mathrm{op},q}\} \leq \eta(x),$$

then

$$\begin{aligned}\operatorname{Hess}_q(G - G_0) &= (1 - \beta)\operatorname{Hess}_q E - E \operatorname{Hess}_q \beta \\ &\quad - d\beta \otimes dE - dE \otimes d\beta, \\ |\operatorname{tr}_P^q \operatorname{Hess}_q(G - G_0)| &\leq 3\eta(x)(1 + 2A_1 + A_2).\end{aligned}$$

Choose the fine approximation so that this last quantity is less than $\tau(x)$. This preserves strict three-convexity.

A sufficiently small value error preserves both signs outside $\operatorname{int}\{\beta = 1\}$, while $G = G_0$ on $\{\beta = 1\}$. Taking also $|G - G_0| < 1$ preserves boundedness below and compact sublevels. Thus G has the same zero sublevel and the same germ near $\partial\Gamma$.

Finally, three negative critical Hessian eigenvalues would contradict the positive three-plane trace, so

$$\operatorname{ind}_x G \le 2 \qquad (x \in \operatorname{Crit} G).$$

□

C.8. **Proof of Theorem 5.12.**

Proof. Retain the fixed potential, metric q, exhaustion F, and regular cores from Theorems 3.3–4.1. Given a compact set $Q \subset M$, choose i such that

$$Q \subset \operatorname{int} C_i.$$

Set $A = C_i$, and take $B = C_j$ and the forcing function k from Proposition 5.3.

Proposition 5.6 gives the weakly outermost domain T, and Corollary 5.7 gives its component K containing A. Its boundary is connected and admits PSC, and inclusion surjects on fundamental groups.

Lemmas 5.8 and C.1 give a connected inward collar cut Γ with

$$Q \subset \operatorname{int} A \subset \operatorname{int} \Gamma, \qquad \Gamma \subset \operatorname{int} K,$$

positive outward mean curvature, and collar data r, c as required in Lemma 5.10. The boundary is diffeomorphic to Y_K, hence admits PSC. Corollary 5.2 gives connectedness and

$$\pi_1(Y_\Gamma) \twoheadrightarrow \pi_1(\Gamma).$$

Since $B = C_j$ is a regular component of $\{F \le s_j\}$, every point of B with $F = s_j$ is a boundary point. Thus $F < s_j$ on Γ. Compactness gives a number b such that

$$\max_\Gamma F < b < s_j = F|_{\partial B}.$$

Lemmas 5.10 and C.4 imply

$$\widehat{\Gamma}_3^q = \Gamma.$$

This also proves Lemma 5.11.

Finally, Lemmas C.5 and C.6 give a smooth strictly three-convex Morse exhaustion G with

$$\Gamma = \{G \le 0\}, \qquad Y_\Gamma = \{G = 0\},$$

and $dG \neq 0$ on Y_Γ. Its critical indices are at most two. The construction keeps q fixed while Q varies and establishes all the asserted properties. □

Appendix D. Proofs of Section 6

All homology groups in this section have integer coefficients unless otherwise specified.

D.1. **Proof of Lemma 6.2.**

Proof. Let a realize $\mathsf{Band}_q(K, B)$, and let D denote either K or B.

Suppose $x \in Y_D$ and $da_x = 0$. Since $a|_{Y_D}$ is constant,

$$(\mathrm{Hess}_q\, a)_x(V, W) = 0 \qquad (V, W \in T_x Y_D).$$

In a frame adapted to $T_x Y_D$, the Hessian has the form

$$\begin{pmatrix} 0_{3\times 3} & \beta \\ \beta^{\mathsf{T}} & \gamma \end{pmatrix},$$

whose rank is at most two. This contradicts the nondegeneracy in Definition 6.1. Hence $da \neq 0$ on both boundaries.

Let ν_q be the outward unit normal of D. At a boundary point,

$$\nabla_q a = (\nu_q a)\nu_q.$$

For tangent vectors $V, W \in TY_D$, we also have

$$\mathrm{Hess}_q\, a(V, W) = (\nu_q a)\mathrm{II}_q(V, W).$$

Therefore

$$0 < \mathrm{tr}^q_{(\nabla_q a)^\perp} \mathrm{Hess}_q\, a = (\nu_q a) H_q(Y_D).$$

Since $H_q(Y_D) > 0$, we have $\nu_q a > 0$. Thus $\nabla_q a$ points outward from both K and B. □

D.2. **Proof of Proposition 6.4.**

Lemma D.1 (Relative handles for a band and its tail)**.** *Suppose $K \subset \operatorname{int} B$ are compact regular domains, $\mathsf{Band}_q(K, B)$ holds, and its boundary gradients point outward from K and B. Let F be a proper, bounded-below Morse function whose critical indices are at most two. Suppose B is a component of a regular sublevel of F.*

Then every finite CW structure on Y_K extends to a relative cell model of (E_K, Y_K) with relative cells only in dimensions $0, 1, 2$. Moreover,

$$\begin{aligned} &H_\ell(M, K) = 0 \quad (\ell \geq 3), \\ &H_2(M, K) \text{ is torsion-free.} \end{aligned} \tag{D.1}$$

Proof. Let a realize $\mathsf{Band}_q(K, B)$, and put

$$W = B \setminus \operatorname{int} K.$$

We first replace $a|_W$ by a Morse function with range $[0, 1]$ and the same critical points and indices.

Choose $0 < \varepsilon < 1/2$ sufficiently small that

$$\frac{1}{4} < \varepsilon a + \frac{1}{2} < \frac{3}{4} \quad \text{on } W.$$

The outward boundary gradients make a a normal coordinate in disjoint boundary collars. On the W-side of Y_K, we have $a > 0$ near the boundary; on the W-side of Y_B, we have $a < 1$.

Shrink the collars so that their a-coordinate ranges are $[0, \delta]$ and $[1 - \delta, 1]$, where $0 < \delta < 1/2$. Choose nonnegative smooth bumps η_-, η_+ supported in the interiors of these intervals, each with integral one. Define

$$r_-(s) = \varepsilon s + \frac{1}{2} \int_0^s \eta_-(t)\, dt,$$
$$r_+(s) = \varepsilon s + \frac{1}{2} + \left(\frac{1}{2} - \varepsilon\right) \int_{1-\delta}^s \eta_+(t)\, dt.$$

Then $r_-(0) = 0$, $r_+(1) = 1$, and

$$r'_- = \varepsilon + \tfrac{1}{2}\eta_- > 0,$$
$$r'_+ = \varepsilon + (\tfrac{1}{2} - \varepsilon)\eta_+ > 0.$$

Near the interior edge of either collar, the corresponding function agrees with $\varepsilon s + 1/2$. Use $r_- \circ a$ and $r_+ \circ a$ in the two collars and $\varepsilon a + 1/2$ elsewhere. The resulting smooth function

$$\widehat{a} : W \longrightarrow [0, 1]$$

satisfies

$$\widehat{a}^{-1}(0) = Y_K, \qquad \widehat{a}^{-1}(1) = Y_B.$$

It introduces no critical points in the collars. Elsewhere it is a positive affine transformation of a, so its critical points and indices agree with those of a.

The relative Morse construction builds W from a collar of Y_K by handles of indices $0, 1, 2$. Each handle contributes a relative cell of its index [24, Theorems 3.13–3.14].

Let s be the regular value defining B, and write

$$\{F \leq s\} = B \sqcup B_1 \sqcup \cdots \sqcup B_\ell.$$

There are finitely many components because the sublevel is a compact smooth manifold with boundary. Each B_α is built from the empty set by the F-handles in that component, again of indices $0, 1, 2$.

Choose regular values above s tending to infinity. Each intervening slab is compact and contains finitely many critical points, so only finitely many handles are attached at each stage. Critical points sharing a critical value can be treated simultaneously [24, remark following Theorem 3.14]. Together with the compact band and the components B_α, these handles construct E_K relative to Y_K.

Start with the prescribed finite CW structure on Y_K. Replace handles by core disks and apply cellular approximation to their attaching maps. Choose the finite-stage models and comparison maps inductively relative to the preceding stage and to Y_K. Their union gives a CW complex X, containing the prescribed Y_K as a subcomplex, and a comparison map

$$e : X \longrightarrow E_K$$

whose restriction to Y_K is homotopic to its inclusion.

Every compact subset of E_K lies in a finite stage, and every compact subset of X lies in a finite subcomplex [16, Proposition A.1]. Consequently, maps of spheres and their homotopies are detected at finite stages. The finite-stage equivalences imply that e is bijective on components and induces isomorphisms on all homotopy groups. Since smooth manifolds have CW type, Whitehead's theorem, applied componentwise, makes e a homotopy equivalence [16, Theorem 4.5].

The homotopy extension property of the CW pair (X, Y_K) allows e to be replaced by a homotopic map satisfying

$$e|_{Y_K} = (Y_K \hookrightarrow E_K)$$

exactly [16, Proposition 0.16]. This gives the relative model of Definition 6.3, including its prescribed boundary cells and actual boundary map.

Collared excision gives

$$H_\ell(M, K) \cong H_\ell(E_K, Y_K).$$

Indeed, excising a slightly smaller core of K leaves a collar pair which deformation retracts to (E_K, Y_K). Since e is a homotopy equivalence and restricts to the actual inclusion of Y_K, the pair sequences give

$$H_\ell(X, Y_K) \cong H_\ell(E_K, Y_K).$$

Relative cellular homology [16, Theorem 2.35 and Section 2.2] computes these groups using a chain complex with

$$C_\ell(X, Y_K) = 0 \quad (\ell \geq 3).$$

Hence

$$H_\ell(M, K) = 0 \quad (\ell \geq 3),$$
$$H_2(M, K) \cong \ker\bigl(\partial_2 : C_2(X, Y_K) \to C_1(X, Y_K)\bigr).$$

The latter is a subgroup of a free abelian group and is therefore torsion-free. □

Proof of Proposition 6.4. Put $B = D_j$ and $W = B \setminus \operatorname{int} K$. The boundary Y_B is strictly mean convex by Theorem 4.1. Together with $H_q(Y_K) > 0$, Lemma 6.2 supplies the outward boundary gradients. Apply Lemma D.1 with tail function $F = H$. It gives the relative cell model and the homology assertions.

To identify the exterior fundamental-group map, read the compact band from Y_B toward Y_K. A forward handle of index $\lambda \leq 2$ becomes a reverse handle of index $4 - \lambda \geq 2$. Thus W is built from a collar of the connected Y_B without relative zero- or one-handles. It follows that W is path connected and

$$\pi_1(Y_B) \twoheadrightarrow \pi_1(W).$$

The closed exteriors satisfy

$$E_K = E_B \cup W, \qquad E_B \cap W = Y_B.$$

Boundary collars give an open cover with these homotopy types. Moreover, $E_B \simeq O_B$ is path connected by Theorem 4.1. Van Kampen [16, Theorem 1.20] gives

$$\pi_1(E_K) \cong \pi_1(E_B) *_{\pi_1(Y_B)} \pi_1(W).$$

Since $\pi_1(Y_B) \to \pi_1(W)$ is surjective, the first vertex group surjects onto this pushout:

$$\pi_1(E_B) \twoheadrightarrow \pi_1(E_K).$$

The commuting square of actual inclusions

$$\begin{array}{ccc} O_B & \longrightarrow & O_K \\ \downarrow & & \downarrow \\ E_B & \longrightarrow & E_K \end{array}$$

has homotopy equivalences as its vertical maps. With compatible basepoints, it follows that

$$\pi_1(O_{D_j}) \twoheadrightarrow \pi_1(O_K)$$

is induced by inclusion. The same decomposition shows that E_K is path connected, and hence so is O_K. □

D.3. **Proof of Lemma 6.6.**

Proof. Contractibility of M implies orientability of M and K. Since K is compact, its homology groups are finitely generated. Thus the torsion-free group $H_1(K)$ is free abelian.

The universal coefficient theorem [16, Theorem 3.2] gives

$$H^2(K) = H^3(K) = 0.$$

Indeed, the relevant terms are

$$\begin{aligned} &\mathrm{Hom}(H_2(K), \mathbb{Z}), \quad \mathrm{Ext}(H_1(K), \mathbb{Z}), \\ &\mathrm{Hom}(H_3(K), \mathbb{Z}), \quad \mathrm{Ext}(H_2(K), \mathbb{Z}), \end{aligned}$$

all of which vanish. Poincaré–Lefschetz duality [16, Theorem 3.43] therefore gives

$$\begin{aligned} H_2(K, Y_K) &\cong H^2(K) = 0, \\ H_1(K, Y_K) &\cong H^3(K) = 0. \end{aligned}$$

The exact sequence of the pair shows that inclusion induces

$$H_1(Y_K) \xrightarrow{\cong} H_1(K).$$

Use boundary collars to apply Mayer–Vietoris to $M = K \cup E_K$. Since M is contractible, the inclusion maps give isomorphisms

$$H_2(Y_K) \xrightarrow{\cong} H_2(K) \oplus H_2(E_K)$$

and

$$H_1(Y_K) \xrightarrow{\cong} H_1(K) \oplus H_1(E_K),$$

up to the usual choice of signs. As $H_2(K) = 0$, the first proves the asserted isomorphism $H_2(Y_K) \to H_2(E_K)$.

In the second, projection to $H_1(K)$ is already an isomorphism. Surjectivity onto an element $(0,\xi)$ therefore forces $\xi = 0$. Hence $H_1(E_K) = 0$. Finally, $H_1(Y_K) \cong H_1(K)$ is torsion-free. □

D.4. **Proof of Proposition 6.7.**

Proof. The simply connected ambient manifold is orientable, so K and $Y = \partial K$ inherit orientations. The PSC classification [7, Proposition 2.2], applied to the identity map of Y, gives

$$Y \cong \left(\#_{\ell=1}^{v} P_\ell \right) \# \left(\#_{i=1}^{r} (S^1 \times S^2) \right),$$

where the P_ℓ are spherical three-manifolds. An empty connected sum is interpreted as S^3.

Connected-sum homology gives

$$H_1(Y) \cong \left(\bigoplus_{\ell=1}^{v} H_1(P_\ell) \right) \oplus \mathbb{Z}^r.$$

Each $H_1(P_\ell)$ is finite. Since $H_1(Y)$ is torsion-free, all these finite summands vanish. Thus each P_ℓ is an integral homology sphere. By [7, Corollary 2.3], it is S^3 or a connected sum of copies of P and $-P$. Its fundamental group is finite, whereas a free product of two nontrivial groups is infinite. It therefore has at most one nontrivial summand. After discarding the S^3 summands, each P_ℓ is P or $-P$. Put

$$Y_0 = \#_{\ell=1}^{v} P_\ell.$$

For each punctured P_ℓ, choose a finite PL spine of dimension at most two. The existence of such spines and their regular-neighborhood description are given in [31, Theorems 2.1 and 2.5]. For each $S^1 \times S^2$ summand, use the product CW structure with embedded two-skeleton

$$(S^1 \times \{p_i\}) \cup (\{t_i\} \times S^2) \cong S^1 \vee S^2.$$

Its complement is an open three-cell:

$$(S^1 \setminus \{t_i\}) \times (S^2 \setminus \{p_i\}) \cong \mathbb{R}^3.$$

The circle contributes one one-cell, and the sphere contributes one two-cell with constant attaching map at the wedge point.

Join the punctured P_ℓ spines and the marked wedge points by a tree of arcs, meeting each spine at a single chosen point. Let A be the union of the P_ℓ spines and this tree. Choose finite CW structures on the spines and tree compatible at these attaching points. The punctured homology spheres are acyclic, and the connecting tree adds no reduced homology. Thus

$$\widetilde{H}_*(A) = 0, \qquad \pi_1(A) \cong *_{\ell=1}^{v} \pi_1(P_\ell).$$

If $v = 0$, take A to be the connecting tree, or a point if no connecting arcs are needed. Every selected sphere cell is attached at a point of A.

Finite groups are residually finite, and free products of residually finite groups are residually finite [3, Section 1.1 and Theorem 1.1]. Hence $\pi_1(A)$ is residually finite.

Take regular neighborhoods of the summand spines and join them by boundary bands along the chosen tree. Their boundary connected sum is the punctured connected sum $Y \setminus \operatorname{int} B^3$, with the joined spine as a spine. The regular-neighborhood description identifies its complement with a boundary collar. Capping by B^3 extends this spine structure to a CW structure on Y with one three-cell. Every two-cell outside A is one of the selected sphere cells; there is one for each $S^1 \times S^2$ summand, so $s = r$. The circle one-cells supply the free generators outside $\pi_1(A)$.

Now choose disjoint loops

$$\ell_i = S^1 \times \{p_i\} \qquad (1 \le i \le r)$$

in the $S^1 \times S^2$ summands, away from the connected-sum balls. Attach four-dimensional two-handles to K along these loops with their product framings. After rounding corners, denote the resulting compact smooth four-manifold by D.

On the boundary, each surgery replaces $S^1 \times D^2$ by $D^2 \times S^1$. The complement of the removed solid torus in its $S^1 \times S^2$ summand is another solid torus, and the product-framed gluing gives

$$(S^1 \times D^2) \cup (D^2 \times S^1) \cong \partial(D^2 \times D^2) \cong S^3.$$

Thus each surgery replaces one $S^1 \times S^2$ summand by S^3, and

$$\partial D \cong Y_0.$$

This is the standard product-framed surgery in [28, Lemma 8.24 and Example 8.26(i)]. The construction requires no spanning disks in the original ambient manifold.

The handle attachment and corner rounding give a continuous embedding $f : K \hookrightarrow D$. Placing D in its smooth double realizes it as a compact regular domain in an auxiliary smooth four-manifold. Its boundary is connected, and

$$\pi_1(\partial D) \cong \pi_1(Y_0) \cong \pi_1(A),$$
$$\pi_1(Y) \cong \pi_1(\partial D) * F_r.$$

Moreover, $H_1(Y) \cong \mathbb{Z}^r$ has basis

$$b_i = [\ell_i] \qquad (1 \le i \le r).$$

The pair (D, K) is obtained by attaching exactly r two-handles. Replacing the handles by their core disks gives

$$H_1(D, K) = 0, \qquad H_2(D, K) \cong \mathbb{Z}^r.$$

Let e_i be the relative class of the oriented core disk of the ith handle. Its boundary is the attaching loop ℓ_i, so the actual connecting homomorphism satisfies

$$\delta e_i = (Y \hookrightarrow K)_*[\ell_i] = (Y \hookrightarrow K)_* b_i.$$

Choose the core-disk orientations consistently with the loops. All the cell, group, and relative-homology identifications therefore arise from the same construction. □

D.5. **Proof of Theorem 6.8.**

Proof. Put $H = \pi_1(A)$. Choose a maximal tree in the one-skeleton of Z/A. Its edges lift to a forest in Z whose components each meet A in exactly one vertex. Contract each lifted tree to that vertex. This fixes A and preserves the homotopy type of the pair, so we may assume there are no relative zero-cells.

Let $x_1, \dots, x_n$ correspond to the remaining relative one-cells, and let $w_1, \dots, w_m$ be the attaching words of the relative two-cells. The relative cellular presentation is

$$\pi_1(Z) \cong \big(H * F(x_1, \dots, x_n)\big)/\langle\langle w_1, \dots, w_m \rangle\rangle.$$

Each w_j has coefficients in H, and the map from H is the original inclusion-induced map [16, Proposition 1.26(a)].

The relative cellular chain complex is

$$0 \longrightarrow \mathbb{Z}^m \xrightarrow{\mathcal{D}} \mathbb{Z}^n \longrightarrow 0.$$

The entry $\mathcal{D}_{ij}$ is the exponent sum of x_i in w_j [16, Theorem 2.35 and Section 2.2]. Since $H_2(Z, A) = 0$, the matrix $\mathcal{D}$ is injective over $\mathbb{Z}$ and has rank m over $\mathbb{Q}$.

If $m = 0$, assign arbitrary unitary matrices to the new generators. Otherwise, choose m variables corresponding to a nonzero $m \times m$ minor of $\mathcal{D}$ and assign the identity to all other variables. Apply ρ to the coefficients in the attaching words. This gives m equations in m variables in $U(d)$ whose exponent-sum determinant is nonzero.

Since $U(d)$ is a compact connected Lie group, [13, Theorem 1(ii)] gives a solution in this same $U(d)$. Equivalently, one may apply [26, Lemma 2.1(1)] directly to the original n-variable system: after erasing the coefficients, its presentation complex has cellular two-boundary $\mathcal{D}$, and hence vanishing H_2. The resulting homomorphism on $H * F(x_1, \dots, x_n)$ annihilates every attaching word and restricts to ρ on H. It therefore descends to

$$\widetilde{\rho} : \pi_1(Z) \longrightarrow U(d), \qquad \widetilde{\rho} \circ i_* = \rho.$$

Suppose now that H is residually finite, and let $1 \neq \gamma \in H$. Choose a homomorphism $b : H \to Q$ to a finite group with $b(\gamma) \neq 1$. Its regular permutation representation is faithful and unitary:

$$\lambda : Q \hookrightarrow U(|Q|).$$

Extend $\lambda \circ b$ by the preceding argument. If $i_*(\gamma) = 1$, the extension would imply $\lambda(b(\gamma)) = I$, a contradiction. Hence i_* is injective. □

D.6. **Proof of Corollary 6.9.**

Proof. Suppose a loop in A is nullhomotopic in Z. Its nullhomotopy has compact image and is therefore contained, together with the finite subcomplex A, in a finite subcomplex $Z_0 \subset Z$ [16, Proposition A.1]. Taking the component containing A, we may assume Z_0 is connected.

Since Z has dimension at most two, cellular chains give

$$H_2(Z_0) \hookrightarrow H_2(Z) = 0.$$

The exact sequence

$$H_2(Z_0) \longrightarrow H_2(Z_0, A) \longrightarrow H_1(A)$$

and $H_1(A) = 0$ imply $H_2(Z_0, A) = 0$. Theorem 6.8 makes $\pi_1(A) \to \pi_1(Z_0)$ injective. Thus the loop is already nullhomotopic in A, proving injectivity into $\pi_1(Z)$. The finiteness assertion follows immediately. □

D.7. **Proof of Lemma 6.10.**

Proof. The selected two-cells have constant attaching maps at points of A. Deleting their open cells from $X^{(2)}$ therefore leaves a subcomplex Z containing $j(A)$. It retains the entire one-skeleton of the connected complex X, so Z is connected.

Write ∂_i^T for the cellular boundary maps of a CW complex T. The closed connected orientable three-manifold Y has one three-cell and no higher cells. Since $H_3(Y) = \mathbb{Z}$, its cellular differential satisfies

$$\partial_3^Y = 0, \qquad H_2(Y^{(2)}) \xrightarrow{\cong} H_2(Y).$$

The relative model adds no cells of dimension at least three. Thus

$$C_3(X) = j_* C_3(Y), \qquad \partial_3^X = 0,$$

and

$$H_2(X^{(2)}) \xrightarrow{\cong} H_2(X).$$

Every two-cell of Y other than the selected sphere cells belongs to A. Since $\dim A \le 2$ and $H_2(A) = 0$, the differential ∂_2^A is injective. The selected cells have zero cellular boundary, so

$$\ker \partial_2^Y = \bigoplus_{\alpha=1}^{s} \mathbb{Z}[e_\alpha^2].$$

The additional one-cells outside A do not affect this calculation: boundaries of two-cells in A lie in $C_1(A)$.

By the defining identity

$$e \circ j = (Y \hookrightarrow E_K)$$

and Lemma 6.6, the map

$$H_2(Y^{(2)}) \xrightarrow{j_*} H_2(X^{(2)})$$

is an isomorphism. Indeed, both skeleton inclusions are isomorphisms in degree two, e is a homotopy equivalence, and the actual inclusion $Y \hookrightarrow E_K$ induces an isomorphism on H_2. As the cellular three-boundaries vanish, this identifies the cycle groups themselves:

$$\ker \partial_2^X = \bigoplus_{\alpha=1}^{s} \mathbb{Z}[j(e_\alpha^2)].$$

On the other hand,

$$C_2(X) = C_2(Z) \oplus \bigoplus_{\alpha=1}^{s} \mathbb{Z}[j(e_\alpha^2)].$$

Each selected cell has zero boundary, and hence

$$\ker \partial_2^X = \ker \partial_2^Z \oplus \bigoplus_{\alpha=1}^{s} \mathbb{Z}[j(e_\alpha^2)].$$

Comparison gives $\ker \partial_2^Z = 0$. Since Z has dimension at most two, $H_2(Z) = 0$.

Adding back two-cells with constant attaching maps does not change the fundamental group. Neither does adding the boundary three-cell. Consequently,

$$\pi_1(Z) \cong \pi_1(X^{(2)}) \cong \pi_1(X) \cong \pi_1(E_K).$$

The equivalence $E_K \simeq O_K$ identifies this group with $\pi_1(O_K)$.

The subcomplex $j(A)$ is finite and connected, with $H_1(j(A)) = 0$ and residually finite fundamental group. Corollary 6.9 gives

$$\pi_1(A) \hookrightarrow \pi_1(Z) \cong \pi_1(E_K) \cong \pi_1(O_K).$$

This is the asserted injection. □

D.8. **Proof of Proposition 6.12.**

Proof. Lemma 6.10 gives an injection

$$\pi_1(A) \hookrightarrow \pi_1(O_K).$$

By Proposition 6.7,

$$\pi_1(A) \cong \mathop{*}_{\ell=1}^{v} I^*, \qquad I^* = \pi_1(P),$$

where I^* is the binary icosahedral group.

We first show that $v \leq 1$ using the group hypothesis in Condition 2. If $v \geq 2$, choose three distinct elements $a_0 = 1, a_1, a_2$ in the first factor and an involution b in the second factor. Such an involution is the central element of order two in I^*. The free-product normal form shows that

$$t_i = a_i b a_i^{-1} \quad (0 \leq i \leq 2)$$

generate a subgroup isomorphic to $(\mathbb{Z}/2) * (\mathbb{Z}/2) * (\mathbb{Z}/2)$: a word in the t_i with unequal adjacent indices expands to a nonempty reduced word in the two original factors. The kernel of the homomorphism sending all three t_i to $1 \in \mathbb{Z}/2$ is free of rank two. Indeed, Schreier rewriting with coset

representatives $1, t_0$ gives the free basis t_0t_1, t_0t_2 and no nontrivial relators. Thus F_2 embeds in $\pi_1(A)$ and hence in $\pi_1(O_K)$, contrary to Condition 2. Therefore $v \leq 1$.

Lemma 6.6 gives an inclusion-induced isomorphism

$$i_* : H_1(Y) \xrightarrow{\cong} H_1(K).$$

For the bases in (6.5), we have $\delta e_i = i_* b_i$. Therefore the actual connecting homomorphism is an isomorphism:

$$\delta : H_2(D, K) \xrightarrow{\cong} H_1(K).$$

The exact sequence

$$\begin{aligned} 0 = H_2(K) \longrightarrow H_2(D) \longrightarrow H_2(D, K) \xrightarrow{\delta} H_1(K) \\ \longrightarrow H_1(D) \longrightarrow H_1(D, K) = 0 \end{aligned}$$

now gives

$$H_1(D) = H_2(D) = 0.$$

The universal coefficient theorem [16, Theorem 3.2] implies

$$H^1(D; \mathbb{Z}/2) = H^2(D; \mathbb{Z}/2) = 0.$$

Hence the obstruction classes

$$w_1(TD) \in H^1(D; \mathbb{Z}/2), \qquad w_2(TD) \in H^2(D; \mathbb{Z}/2)$$

vanish. Thus D is orientable and spin. Moreover, $H_2(D; \mathbb{R}) = 0$, so $\sigma(D) = 0$.

If $v = 1$, then $\partial D \cong P$ or $-P$. Theorem 6.11 would give

$$0 = \sigma(D) \equiv 8\mu(\pm P) = 8 \pmod{16},$$

a contradiction. Therefore $v = 0$ and $\partial D \cong S^3$. Consequently, $\pi_1(A) \cong \pi_1(\partial D) = 1$, and

$$\pi_1(Y) \cong \pi_1(\partial D) * F_r \cong F_r.$$

□

D.9. **Proof of Lemma 6.13.**

Proof. Put $N = \ker a$. Injectivity of $H_1(a; \mathbb{Z})$ implies that every element of N has trivial image in the abelianization of F_r. Thus

$$N \subset [F_r, F_r].$$

Hopf's formula [4, Section 0] gives

$$H_2(G; \mathbb{Z}) \cong \frac{N \cap [F_r, F_r]}{[F_r, N]} = \frac{N}{[F_r, N]}.$$

Since $H_2(G; \mathbb{Z}) = 0$, it follows that

$$N = [F_r, N].$$

Define the lower central series by

$$\gamma_1(F_r) = F_r, \qquad \gamma_{\ell+1}(F_r) = [F_r, \gamma_\ell(F_r)].$$

Inductively,

$$N \subset \gamma_\ell(F_r) \quad \Longrightarrow \quad N = [F_r, N] \subset \gamma_{\ell+1}(F_r).$$

Hence

$$N \subset \bigcap_{\ell \geq 1} \gamma_\ell(F_r) = 1,$$

where the last equality is residual nilpotence of free groups [22, Corollary 1.4]. Thus a is injective and therefore an isomorphism. □

Appendix E. Proofs of Section 7

E.1. **Proof of Proposition 7.1.**

Proof. Apply Theorem 4.1 to $H = G$, and write the resulting regular levels and cores as t_i and D_i. Their interiors exhaust M, so compactness of Γ allows us to choose j with

$$\Gamma \subset \operatorname{int} D_j, \qquad t_j > 0.$$

Put $B = D_j$, $s = t_j$, and $a = G/s$. Since

$$\Gamma = \{G \leq 0\}, \qquad G|_{Y_\Gamma} = 0, \qquad G|_{Y_B} = s,$$

the function a has the required boundary values.

Positive rescaling preserves critical points, nondegeneracy, and Morse indices. At a critical point x,

$$\operatorname{Hess}_q a = s^{-1} \operatorname{Hess}_q G, \qquad \operatorname{ind}_x a = \operatorname{ind}_x G \leq 2.$$

At every regular point of the band,

$$\operatorname{tr}^q_{(\nabla_q a)^\perp} \operatorname{Hess}_q a = s^{-1} \operatorname{tr}^q_{(\nabla_q G)^\perp} \operatorname{Hess}_q G \geq s^{-1} \lambda^q_{(3)}(\operatorname{Hess}_q G) > 0.$$

Thus a realizes $\mathsf{Band}_q(\Gamma, B)$.

Theorem 4.1 gives connected Y_B, path-connected O_B, positive outward mean curvature, and finite $\pi_1(O_B)$. Theorem 5.12 supplies the required properties of Γ, including $H_q(Y_\Gamma) > 0$. Apply Proposition 6.4 and Corollary 6.5 with

$$H = G, \qquad K = \Gamma, \qquad D_j = B.$$

They give a relative cell model for every prescribed finite CW structure on Y_Γ and show that O_Γ is path connected, with

$$|\pi_1(O_\Gamma)| < \infty.$$

They also give

$$H_2(\Gamma) = H_3(\Gamma) = 0,$$

and show that $H_1(\Gamma)$ is torsion-free. In particular, $\pi_1(O_\Gamma)$ contains no subgroup isomorphic to F_2, and the relative model is available for the boundary CW structure chosen in Proposition 6.7. The PSC boundary and inclusion-induced surjection $\pi_1(Y_\Gamma) \twoheadrightarrow \pi_1(\Gamma)$ from Theorem 5.12 supply the remaining clauses of Condition 2. □